



Chapter 3

Random Variables and Probability Distributions





Random Variables and Probability Distributions

- ♣ Random Variables and Probability

- ♣ Discrete Random Variables

 - ◆ Binomial Distribution

 - ◆ Poisson Process

- ♣ Continuous Random Variables

- ♣ Normal Approximation to the Binomial and Poisson Distributions

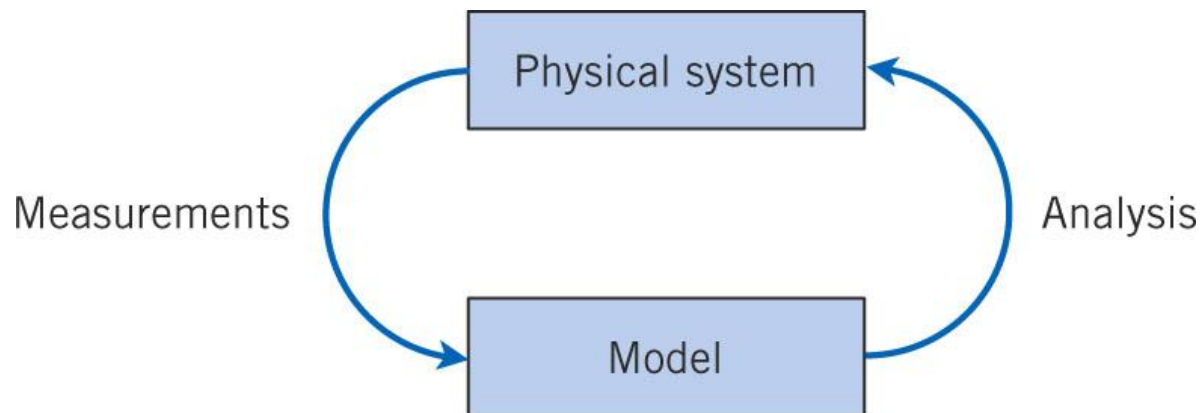
- ♣ More than One Random Variable and Independence

- ♣ Functions of Random Variables



Random Experiments

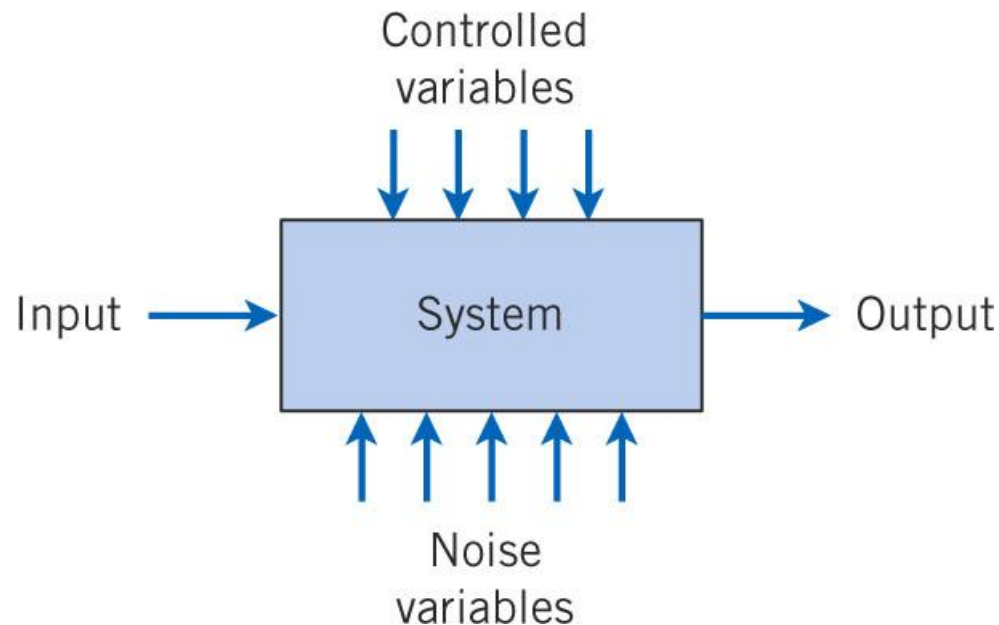
- ♣ The goal is to understand, quantify, and model the variation affecting a physical system's behaviour.
- ♣ The model is used to analyse and predict the physical system's behaviour as system inputs affect system outputs.
- ♣ The predictions are verified through experimentation with the physical system.



Continuous iteration between model and physical system.

Noise Produces Output Variation

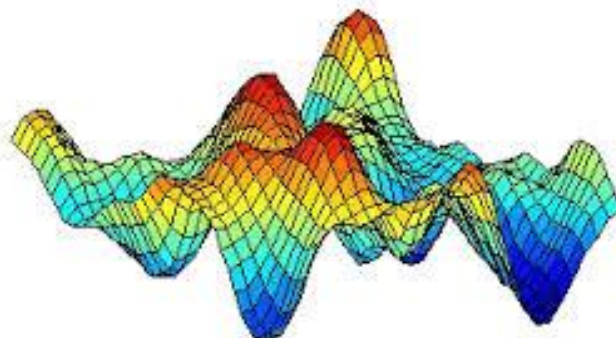
- ♣ Random values of the noise variables **cannot be controlled** and cause the random variation in the output variables.
- ♣ Holding the controlled inputs constant does not keep the output values **constant**.



Noise variables affect the transformation of inputs to outputs.

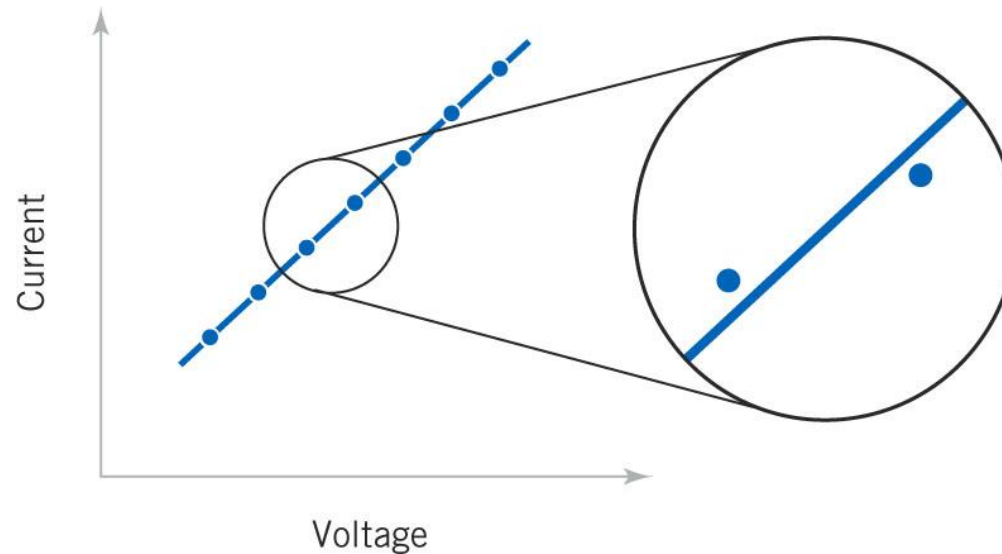
Random Experiment

- ♣ An **experiment** is an operation or procedure, carried out **under controlled conditions**, executed to discover an unknown result or to illustrate a known law.
- ♣ An experiment that can result in different outcomes, even if repeated in the same manner every time, is called a **random experiment**.



Randomness Affects Natural Law

- ♣ **Ohm's Law:** Current is a linear function of voltage.
- ♣ However, current will vary due to noise variables, even under constant voltage.



A closer examination of the system identifies deviations from the model.



Sample Spaces

- ♣ Random experiments have unique **outcomes**.
- ♣ The set of all possible outcome of a random experiment is called the **sample space**, S .
 - ◆ S is **discrete** if it consists of a finite or countable infinite set of outcomes.
 - ◆ S is **continuous** if it contains an interval (either a finite or infinite width) of real numbers.

Example: Defining Sample Spaces

- ♣ Randomly select and measure the thickness of a part.
 - ◆ $S = R^+ = \{x \mid x > 0\}$

The positive real line. Negative or zero thickness is not possible.
 - ◆ S is continuous.
- ♣ It is known that the thickness is between 10 and 11 mm.
 - ◆ $S = \{x \mid 10 < x < 11\}$
 - ◆ S is continuous.
- ♣ It is known that the thickness has only three values.
 - ◆ $S = \{low, medium, high\}$
 - ◆ S is discrete.
- ♣ Does the part thickness meet specifications?
 - ◆ $S = \{yes, no\}$
 - ◆ S is discrete.



Example: Defining Sample Spaces, $n=2$

- ♣ Two parts are randomly selected & measured.
 - ◆ $S = R^+ \times R^+$,
 - ◆ S is continuous.
- ♣ Do the 2 parts conform to specifications?
 - ◆ $S = \{yy, yn, ny, nn\}$,
 - ◆ S is discrete.
- ♣ Number of conforming parts?
 - ◆ $S = \{0, 1, 2\}$,
 - ◆ S is discrete.
- ♣ Parts are randomly selected until a non-conforming part is found.
 - ◆ $S = \{n, yn, yyn, yyyn, \dots\}$,
 - ◆ S is countably infinite.





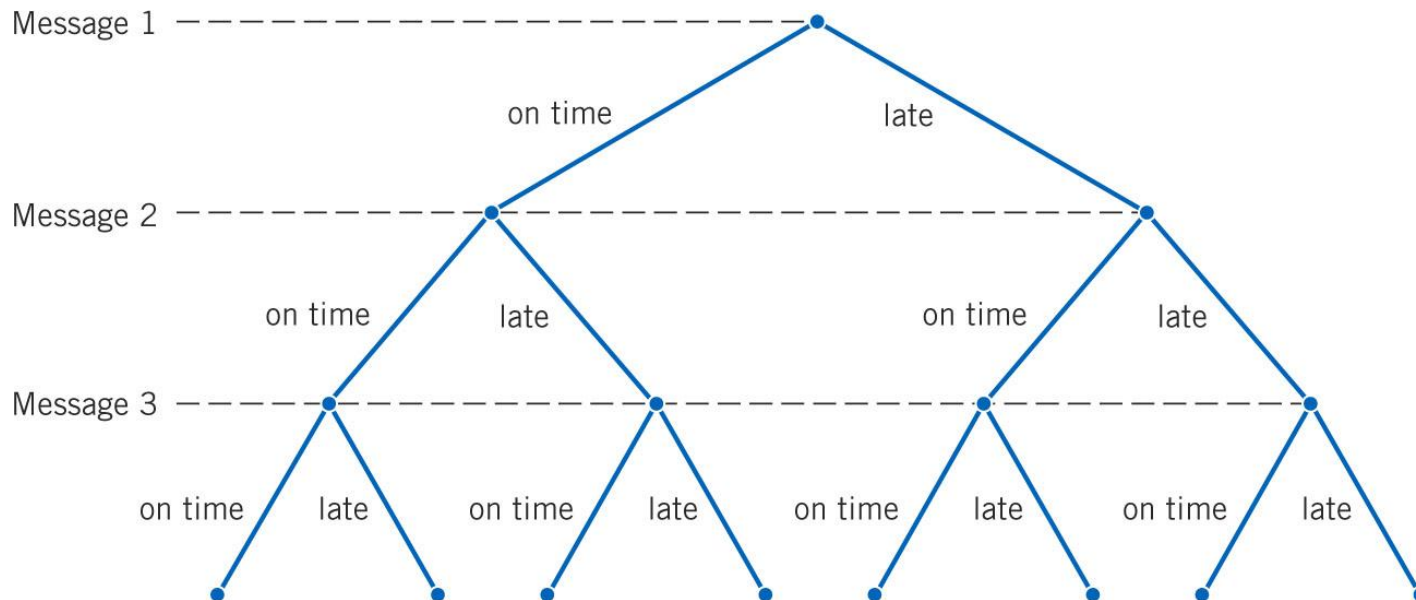
Exercise-1

Decide whether a discrete or continuous random variable is the best model for each of the following variables.

1. The lifetime of a biomedical device after implant in a patient.
2. The number of times a transistor in a computer memory changes state in a time interval.
3. The energy generated from a reaction.
4. The concentration of organic solids in a water sample.
5. The number of convenience options selected by an automobile buyer.

Sample Space Is Defined By A Tree Diagram

- ♣ Example: Messages are classified as on-time or late. 3 messages are classified. There are $2^3 = 8$ outcomes in the sample space.
- ♣ $S = \{ooo, ool, olo, oll, loo, lol, llo, lll\}$

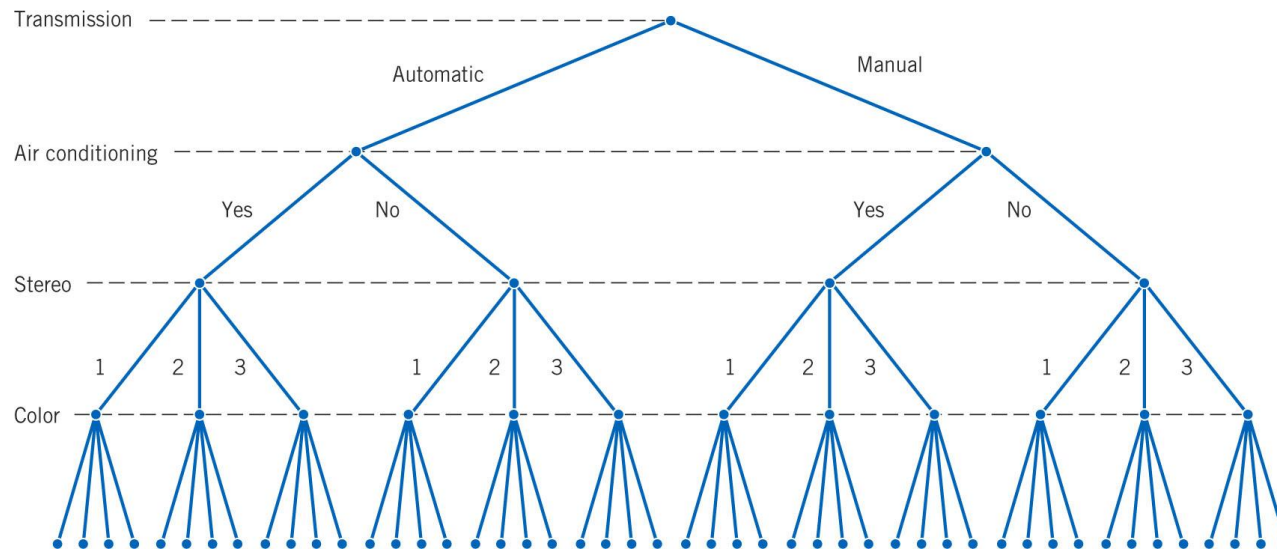
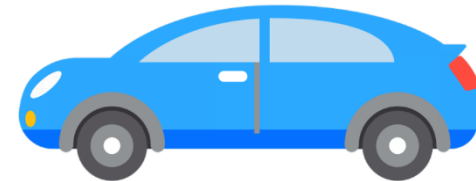


Tree diagram for three messages.

Tree Diagrams Can Fit The Situation

♣ Example: New cars can be equipped with selected options as follows:

1. Manual or automatic transmission
2. With or without air conditioning
3. Three choices of stereo sound systems
4. Four exterior color choices



Tree diagram for different configurations of vehicles.

Note that S has $2 \times 2 \times 3 \times 4 = 48$ outcomes.



Events Are Sets of Outcomes

- ♣ An **event** (E) is a subset of the sample space of a random experiment, i.e., one or more outcomes of the sample space.
- ♣ Event combinations are:
 - ◆ **Union** of two events is the event consisting of all outcomes that are contained in **either** of two events, $E_1 \cup E_2$.
Called E_1 or E_2 .
 - ◆ **Intersection** of two events is the event consisting of all outcomes that contained in **both** of two events, $E_1 \cap E_2$. Called E_1 and E_2 .
 - ◆ **Complement** of an event is the set of outcomes that are **not** contained in the event, E' or not E .



Example: Discrete Event Algebra

Recall the sample space $S = \{yy, yn, ny, nn\}$ concerning conformance to specifications.

- ♣ Let E_1 denote the event that at least one part does conform to specifications, $E_1 = \{yy, yn, ny\}$
- ♣ Let E_2 denote the event that no part conforms to specifications, $E_2 = \{nn\}$
- ♣ Let $E_3 = \emptyset$, the null or empty set.
- ♣ Let $E_4 = S$, the universal set.
- ♣ Let $E_5 = \{yn, ny, nn\}$, at least one part does not conform.

1. Then $E_1 \cup E_5 =$

2. Then $E_1 \cap E_5 =$

3. Then $E_1' =$



Example: Continuous Event Algebra

♣ Measurements of the thickness of a part are modelled with the sample space: $S = \mathbb{R}^+$.

◆ Let $E_1 = \{x \mid 10 \leq x < 12\}$

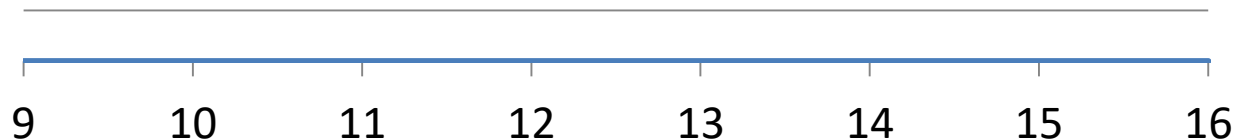
◆ Let $E_2 = \{x \mid 11 < x < 15\}$

1. Then $E_1 \cup E_2 = \{x \mid 10 \leq x < 15\}$

2. Then $E_1 \cap E_2 =$

3. Then $E_1' =$

4. Then $E_1' \cap E_2 =$





Example: Hospital Emergency Visits

- ♣ This table summarizes the ER visits at 4 hospitals. People may leave without being seen by a physician (LWBS). The remaining people are seen, and may or may not be admitted.

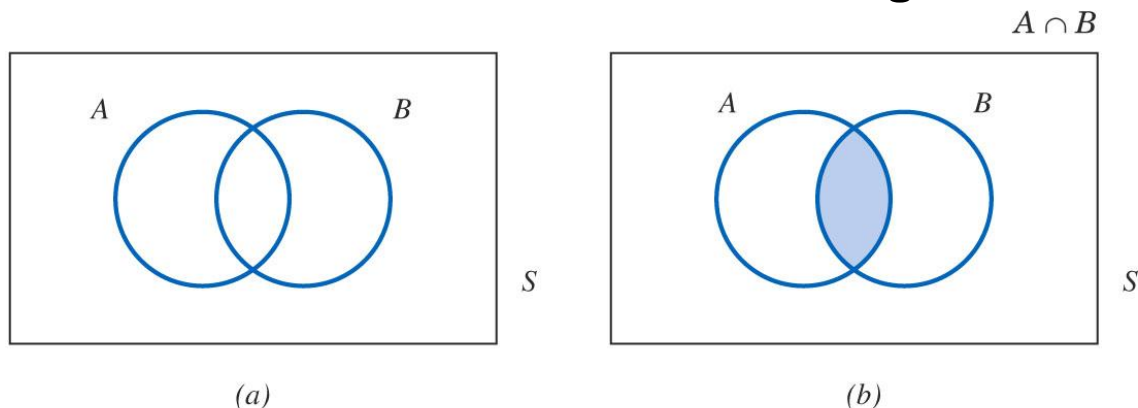
	Hospital				
	1	2	3	4	Total
Total	5,292	6,991	5,640	4,329	22,252
LWBS	195	270	246	242	953
Admitted	1,277	1,558	666	984	4,485
Not admitted	3,820	5,163	4,728	3,103	16,814

- ♣ Let A be the event of a visit to Hospital 1.
- ♣ Let B be the event that the visit is LWBS.
- ♣ Find number of outcomes in:
 - ◆ $A \cap B$
 - ◆ A'
 - ◆ $A \cup B$

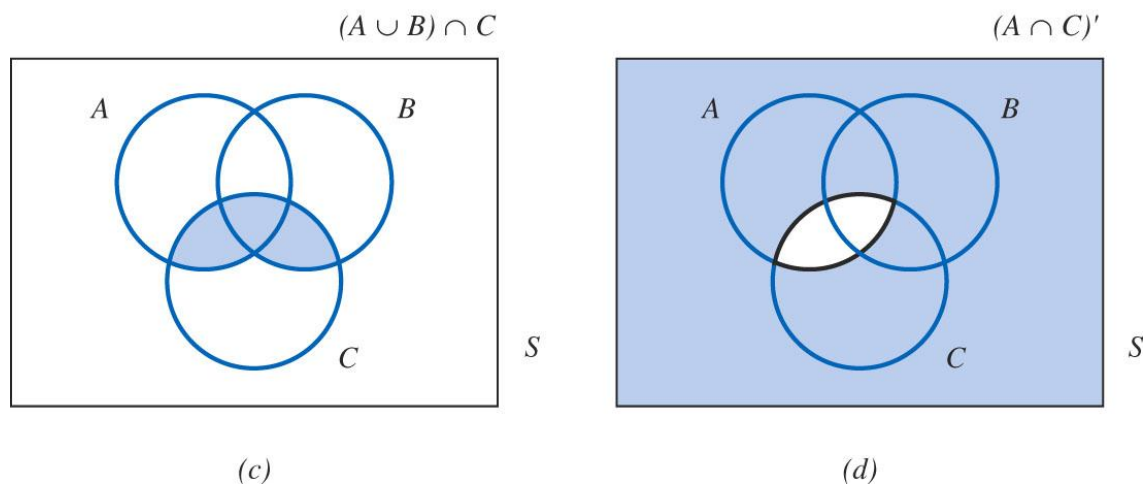


Venn Diagrams Show Event Relations

- ♣ Events A & B contain their respective outcomes. The shaded regions indicate the event relation of each diagram.



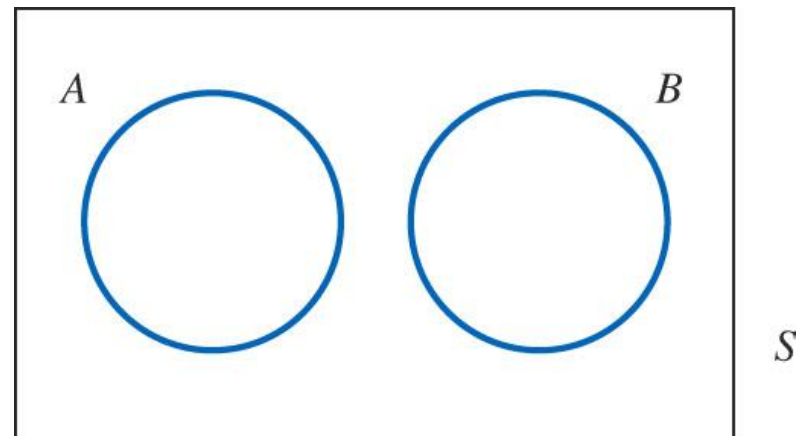
Sample space S with events A and B





Venn Diagram of Mutually Exclusive Events

- Events A & B are mutually exclusive because they share **no common outcomes**.
- The occurrence of one event precludes the occurrence of the other.
- Symbolically, $A \cap B = \emptyset$



Mutually exclusive events



Event Relation Laws

♣ Transitive law (event order is unimportant):

◆ $A \cap B = B \cap A$ and $A \cup B = B \cup A$

♣ Distributive law (like in algebra):

◆ $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$

◆ $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

♣ DeMorgan's laws:

◆ $(A \cup B)' = A' \cap B'$ The complement of the union is the intersection of the complements.

◆ $(A \cap B)' = A' \cup B'$ The complement of the intersection is the union of the complements.



Counting Techniques

- ♣ There are three special rules, or counting techniques, used to determine the number of outcomes in the events and the sample space.
- ♣ They are the:
 - ♦ Multiplication rule
 - ♦ Permutation rule
 - ♦ Combination rule
- ♣ Each has its special purpose that must be applied properly – the right tool for the right job.

Multiplication Rule

- ♣ Let an operation consist of k steps and
 - n_1 ways of completing step 1,
 - n_2 ways of completing step 2, ... and
 - n_k ways of completing step k .
- ♣ Then, the total number of ways or outcomes are: $n_1 * n_2 * \dots * n_k$
- ♣ **Example:** In the design for a gear housing, we can choose to use among:
 - 4 different fasteners,
 - 3 different bolt lengths and
 - 2 different bolt locations.
- ♣ How many designs are possible?





Permutation Rule

- ♣ A permutation is a unique sequence of distinct items.
- ♣ If $S = \{a, b, c\}$, then there are 6 permutations
 - ◆ Namely: abc, acb, bac, bca, cab, cba (**order matters**)
 - ◆ The number of ways 3 people can be arranged.
- ♣ Number of permutations for a set of **n items** is **n!**
 - ◆ $n!$ (factorial function) = $n * (n-1) * (n-2) * \dots * 2 * 1$
 - ◆ $7! = 7 * 6 * 5 * 4 * 3 * 2 * 1 = 5,040$
- ♣ By definition: $0! = 1$





Subset Permutations

- ♣ To sequence only r items from a set of n items:

$$P_r^n = n(n-1)(n-2)\dots(n-r+1) = \frac{n!}{(n-r)!}$$

$$P_3^7 = \frac{7!}{(7-3)!} = \frac{7!}{4!} = \frac{7*6*5*4!}{4!} = 7*6*5 = 210$$

- ♣ **Example:** Circuit Board Designs
- ♣ A printed circuit board has eight different locations in which a component can be placed. If four different components are to be placed on the board, how many designs are possible?
- ♣ **Answer:** order is important, so use the permutation formula with $n = 8$, $r = 4$.

$$P_4^8 =$$

Similar Item Permutations

- ♣ Used for counting the sequences when not all the items are different.
- ♣ The number of permutations of $n = n_1 + n_2 + \dots + n_r$ items of which
 - n_1 are identical,
 - n_2 are identical, ... , and
 - n_r are identical.
- ♣ Is calculated as:
$$\frac{n!}{n_1! n_2! \dots n_r!}$$
- ♣ **Example:** Bar Codes
- ♣ A part is labeled with 4 thick lines, 3 medium lines, and two thin lines. Each sequence is a different label.
- ♣ How many unique labels can be created?



BARCODE 1 2 3 4 5



Combination Rule

- ♣ A combination is a selection of r items from a set of n where **order does not matter**.
- ♣ If $S = \{a, b, c\}$, $n = 3$, then
 - ♦ If $r = 3$, there is 1 combination, namely: abc
 - ♦ If $r = 2$, there are 3 combinations, namely ab , ac , bc
- ♣ # of permutations $\geq$ # of combinations

$$C_r^n = \frac{n!}{r!(n-r)!}$$

- ♣ **Example:** A circuit board has eight locations in which a component can be placed. If 5 identical components are to be placed on a board, how many different designs are possible?
- ♣ **Answer:** The order of the components is not important, so the combination rule is appropriate.



What Is Probability?

- ♣ **Probability** is the likelihood or chance that a particular outcome or event from a random experiment will occur.
- ♣ Here, only **finite sample spaces** ideas apply.
- ♣ Probability is a number in the **[0,1] interval**.
- ♣ May be expressed as a:
 - proportion (0.15)
 - percent (15%)
 - fraction (3/20)
- A probability of:
 - 1 means certainty
 - 0 means impossibility

Probability Based on Equally-Likely Outcomes

- ♣ Whenever a sample space consists of N possible outcomes that are equally likely, the probability of each outcome is $1/N$.

- ♣ **Example:** In a batch of 100 diodes, 1 diode is colored red.
 - ◆ A diode is randomly selected from the batch.
 - ◆ Random means each diode has an equal chance of being selected.
 - ◆ The probability of choosing the red diode is $1/100$ or 0.01 , because each outcome in the sample space is equally likely.





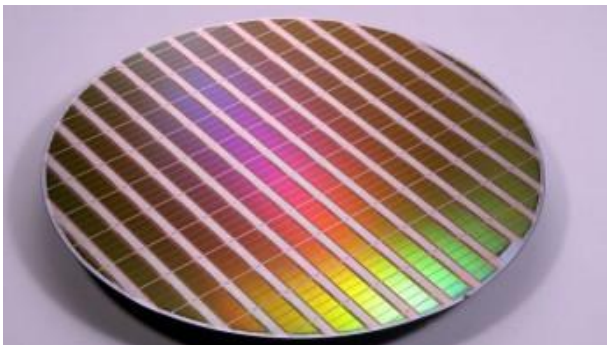
Probability of an Event

- ♣ For a discrete sample space, the *probability of an event E* , denoted by $P(E)$, equals the sum of the probabilities of the outcomes in E .
- ♣ The discrete sample space may be:
 - ◆ A finite set of outcomes
 - ◆ A countably infinite set of outcomes.

Example: Contamination Particles

- ♣ An inspection of a large number of semiconductor wafers revealed the data for this table. A wafer is selected randomly.

- Let E_1 be the event of selecting a 0 particle wafer. $P(E_1) =$
- Let E_2 be the event of selecting a wafer with 3 or more particles. $P(E_2) =$



Number of Contamination Particles	Proportion of Wafers
0	0.40
1	0.20
2	0.15
3	0.10
4	0.05
5 or more	0.10
Total	1.00



Life of Semiconductor

- ♣ Let X denotes the life of a semiconductor laser (in hours) with the following probabilities: $P(X \leq 6000) = 0.05$ and $P(X > 8000) = 0.45$
 - ◆ What is the probability that the life is less than or equal to 8000 hours?
 - ◆ What is the probability that the life is greater than 6000 hours?
 - ◆ What is $P(6000 < X \leq 8000)$?



Life of Semiconductor

- ♣ Let X denotes the life of a semiconductor laser (in hours) with the following probabilities: $P(X \leq 6000) = 0.05$ and $P(X > 8000) = 0.45$
 - ◆ What is the probability that the life is less than or equal to 8000 hours?
 - ♥ $P(X \leq 8000) = 0.55$
 - ◆ What is the probability that the life is greater than 6000 hours?
 - ♥ $P(X > 6000) = 0.95$
 - ◆ What is $P(6000 < X \leq 8000)$?
 - ♥
$$\begin{aligned} P(6000 < X \leq 8000) &= P(X \leq 8000) - P(X \leq 6000) \\ &= 0.55 - 0.05 \\ &= 0.50 \end{aligned}$$



Exercise-2: Probabilities of Events

- ♣ A random experiment has a sample space $\{w,x,y,z\}$.
- ♣ These outcomes are not equally-likely; their probabilities are:
0.1, 0.3, 0.5, 0.1, respectively.
- ♣ Event $A = \{w,x\}$, event $B = \{x,y,z\}$, event $C = \{z\}$
 1. $P(A) = 0.1 + 0.3 = 0.4$
 2. $P(B) = ??$
 3. $P(C) = ??$
 4. $P(A') = 0.6$ and $P(B') = ??$ and $P(C') = ??$
 5. Since event $A \cap B = \{x\}$, then $P(A \cap B) = ??$
 6. Since event $A \cup B = \{w,x,y,z\}$, then $P(A \cup B) = ??$
 7. Since event $A \cap C = \{\text{null}\}$, then $P(A \cap C) = ??$
 8. $P((A \cap B) \cup C) = ??$



Random Variables and Probability Distributions

- ♣ Random Variables and Probability
- ♣ Discrete Random Variables
 - ◆ Binomial Distribution
 - ◆ Poisson Process
- ♣ Continuous Random Variables
- ♣ Normal Approximation to the Binomial and Poisson Distributions
- ♣ More than One Random Variable and Independence
- ♣ Functions of Random Variables





Random Variables

- ♣ A variable that associates a number with the outcome of a random experiment is called a **random variable**.
- ♣ A **random variable** is a function that assigns a real number to each outcome in the sample space of a random experiment.
- ♣ Particular notation is used to distinguish the random variable (rv) from the real number.
 - ◆ The rv is denoted by an uppercase letter, such as **X** .
 - ◆ After the experiment is conducted, the measured value is denoted by a lowercase letter, such as **x is 70**.
 - ◆ X and x are shown in italics, e.g., $P(X=x)$.



Continuous & Discrete Random Variables

- ♣ A **discrete** random variable is a rv with a finite (or countably infinite) range.
 - ◆ They are usually integer counts, e.g., number of errors or number of bit errors per 100,000 transmitted (rate).
 - ◆ The ends of the range of rv values may be finite ($0 \leq x \leq 5$) or infinite ($x \geq 0$).

- ♣ A **continuous** random variable is a rv with an interval (either finite or infinite) of real numbers for its range.
 - ◆ Its precision depends on the measuring instrument.



Examples of Discrete & Continuous RVs

♣ Discrete rv's:

- ◆ Number of scratches on a surface.
- ◆ Proportion of defective parts (count) among 100 tested.
- ◆ Number of transmitted bits received in error.
- ◆ Number of common stock shares traded per day.

♣ Continuous rv's:

- ◆ Electrical current and voltage.
- ◆ Physical measurements, e.g., length, weight, time, temperature, pressure.

Discrete Random Variables

- ♣ **Example:** Voice Lines
- ♣ A voice communication system for a business contains **48 external lines**. At a particular time, the system is observed, and some of the lines are being used.
- ♣ Let **X denote the number of lines in use**. Then, X can assume any of the integer values 0 through 48.
- ♣ The system is observed at a random point in time. If 10 lines are in use, then $x = 10$.





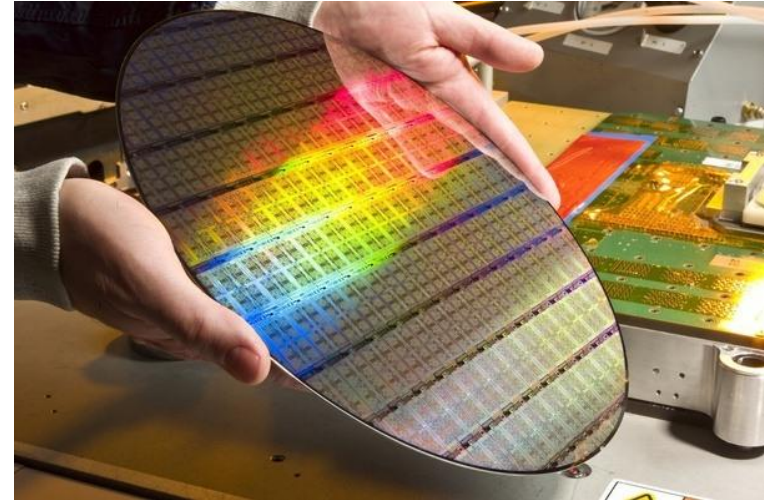
Discrete Random Variables

- ♣ **Example:** Wafers
- ♣ In a semiconductor manufacturing process, **2 wafers** from a lot are sampled. Each wafer is classified as **pass or fail**. Assume that the probability that a wafer passes is 0.8, and that wafers are independent.
- ♣ The sample space for the experiment and associated probabilities are shown in the table. The probability that the 1st wafer passes and the 2nd fails, denoted as pf is
$$P(pf) = 0.8 * 0.2 = 0.16.$$
- ♣ The random variable X is defined as the number of wafers that pass.

Wafer Tests			
Outcome			
Wafer #			
1	2	Probability	x
Pass	Pass	0.64	2
Fail	Pass	0.16	1
Pass	Fail	0.16	1
Fail	Fail	0.04	0
		1.00	

Discrete Random Variables

- ♣ **Example:** Particles on Wafers
- ♣ Define the random variable X to be the number of contamination particles on a wafer.
- ♣ Although wafers possess a number of characteristics, the random variable X summarizes the wafer only in terms of the number of particles.
- ♣ The possible values of X are the integers 0 through a very large number, so we write $x \geq 0$.





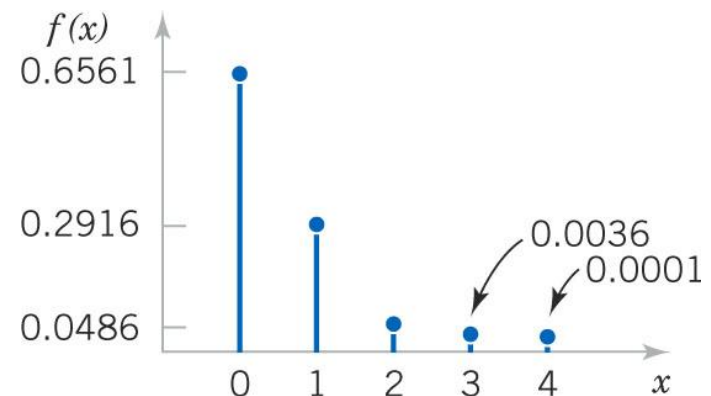
Probability Distributions

- ♣ A **random variable X** associates the **outcomes of a random experiment** to a number on the number line.
- ♣ The **probability distribution** of the random variable X is a description of the probabilities with the possible numerical values of X .
- ♣ A probability distribution of a discrete random variable can be:
 1. A **list of the possible values** along with their probabilities.
 2. A formula that is used to calculate the probability in response to an input of the random variable's value.

Example: Digital Channel

- ♣ There is a chance that a bit transmitted through a digital transmission channel is received in error.
- ♣ Let X equal the number of bits received in error of the next 4 transmitted.
- ♣ The associated probability distribution of X is shown as a graph and as a table.

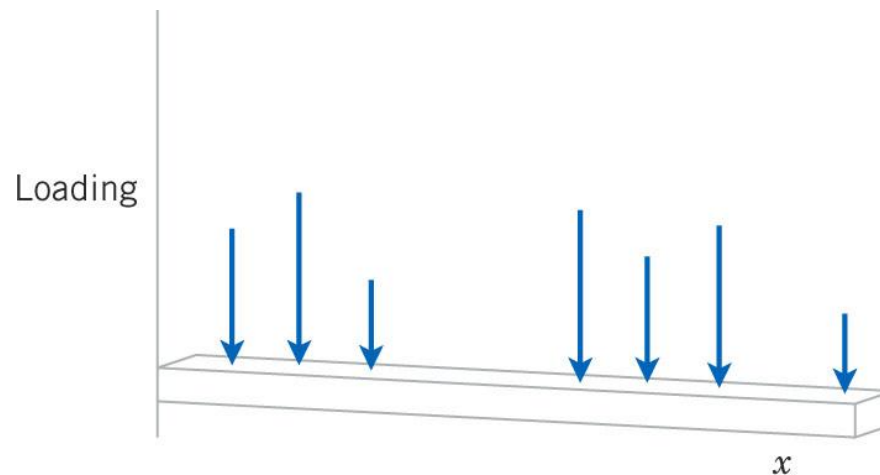
Probability distribution for bits in error.



$P(X=0) =$	0.6561
$P(X=1) =$	0.2916
$P(X=2) =$	0.0486
$P(X=3) =$	0.0036
$P(X=4) =$	0.0001
	1.0000

Probability Mass Function

- ♣ Suppose a loading on a long, thin beam places mass only at discrete points.
- ♣ This represents a **probability distribution** where the beam is the number line over the range of x and the probabilities represent the mass.
- ♣ That's why it is called a **probability mass function**.



Loading at discrete points on a long, thin beam.



Probability Mass Function Properties

For a discrete random variable X with possible values $x_1, x_2, \dots, x_n$, a **probability mass function** is a function such that:

$$(1) \quad f(x_i) \geq 0$$

$$(2) \quad \sum_{i=1}^n f(x_i) = 1$$

$$(3) \quad f(x_i) = P(X = x_i)$$



Example: Wafer Contamination

- ♣ Let the random variable X denote the number of wafers that need to be analyzed to detect a large particle. Assume that the probability that a wafer contains a large particle is 0.1, and that the wafers are independent. Determine the probability distribution of X .
- ♣ Let p denote a wafer for which a large particle is present & let a denote a wafer in which it is absent.
- ♣ The sample space is:
 $S = \{p, ap, aap, aaap, \dots\}$
- ♣ The range of the values of X is: $x = 1, 2, 3, 4, \dots$

Probability Distribution		
$P(X=1) =$	0.1	0.1
$P(X=2) =$	$(0.9)*0.1$	0.09
$P(X=3) =$	$(0.9)^2*0.1$	0.081
$P(X=4) =$	$(0.9)^3*0.1$	0.0729
		0.3439



Cumulative Distribution Function Properties

- ♣ The **cumulative distribution function** is built from the probability mass function and vice versa.

The cumulative distribution function of a discrete random variable X , denoted as $F(x)$

For a discrete random variable X , $F(x)$ satisfies the following properties:

$$(1) F(x) = P(X \leq x) = \sum_{x_i \leq x} f(x_i)$$

$$(2) 0 \leq F(x) \leq 1$$

$$(3) \text{ If } x \leq y, \text{ then } F(x) \leq F(y)$$



Cumulative Distribution Functions

- ♣ We can express the probability of three or fewer bits being in error, denoted as $P(X \leq 3)$.
- ♣ The event $(X \leq 3)$ is the union of the **mutually exclusive** events: $(X=0)$, $(X=1)$, $(X=2)$, $(X=3)$.

x	$P(X=x)$	$P(X \leq x)$
0	0.6561	0.6561
1	0.2916	0.9477
2	0.0486	0.9963
3	0.0036	0.9999
4	0.0001	1.0000
	1.0000	

- ♣ From the table:

$$P(X \leq 3) = P(X=0) + P(X=1) + P(X=2) + P(X=3) = 0.9999$$

$$P(X = 3) = P(X \leq 3) - P(X \leq 2) = 0.0036$$

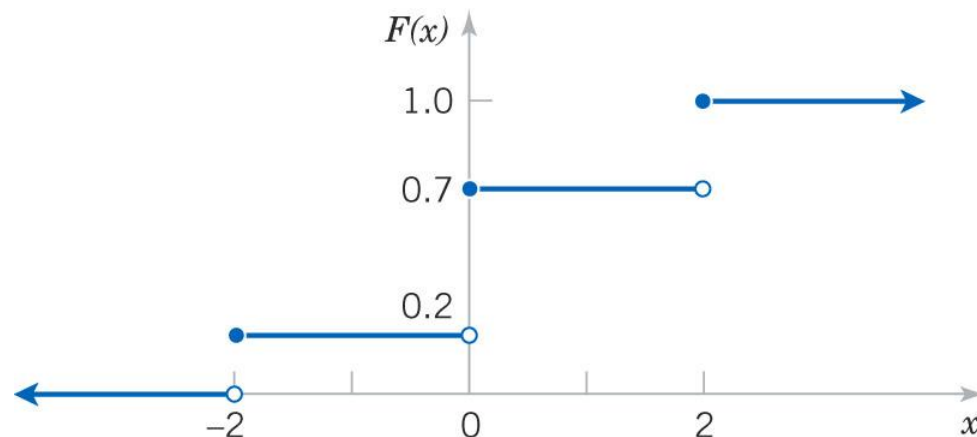


Example: Cumulative Distribution Function

- ♣ Determine the probability mass function of X from this cumulative distribution function:

PMF	
$f(-2) =$	0.2
$f(0) =$	0.5
$f(2) =$	0.3

$F(x) =$	0.0	$x < -2$
	0.2	$-2 \leq x < 0$
	0.7	$0 \leq x < 2$
	1.0	$2 \leq x$



Graph of the CDF



Exercise: Marketing

- ♣ Two new product designs are to be compared on the basis of revenue potential.
 - ♦ Revenue from Design A is predicted to be \$3 million.
 - ♦ But for Design B, the revenue could be \$7 million with probability 0.3 or only \$2 million with probability 0.7.
- ♣ Which design is preferable?



Summary Numbers of a Probability Distribution

- ♣ The **mean** is a measure of the center of a probability distribution.
- ♣ The **variance** is a measure of the dispersion or variability of a probability distribution.
- ♣ The **standard deviation** is another measure of the dispersion. It is the square root of the variance.



Mean Defined

The **mean** or **expected value** of the discrete random variable X , denoted as μ or $E(X)$, is

$$\mu = E(X) = \sum_x x \cdot f(x)$$

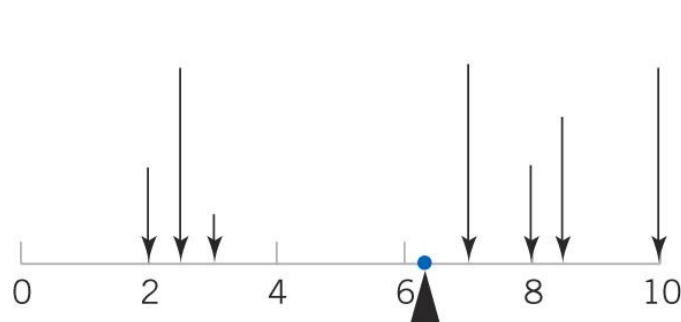
- ♣ The mean is the **weighted average of the possible values of X** , the weights being the probabilities where the beam balances. It represents the center of the distribution. It is also called the **arithmetic mean**.
- ♣ If $f(x)$ is the probability mass function representing the loading on a long, thin beam, then $E(X)$ is the fulcrum or point of balance for the beam.
- ♣ The mean value may, or may not, be a given value of x .

Variance Defined

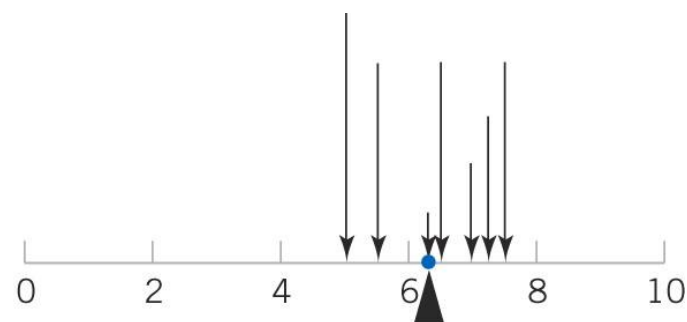
The **variance** of X , denoted as σ^2 or $V(X)$, is

$$\sigma^2 = V(X) = E(X - \mu)^2 = \sum_x (x - \mu)^2 \cdot f(x) = \sum_x x^2 \cdot f(x) - \mu^2$$

- ♣ The variance is the measure of **dispersion or scatter** in the possible values for X .
- ♣ It is the average of the squared deviations from the distribution mean.



(a)



(b)

The mean is the balance point.

Distributions (a) & (b) have equal mean, but (a) has a larger variance.



Variance Formula Derivations

$$\begin{aligned} V(X) &= \sum_x (x - \mu)^2 f(x) \text{ is the definitional formula} \\ &= \sum_x (x^2 - 2\mu x + \mu^2) f(x) \\ &= \sum_x x^2 f(x) - 2\mu \sum_x x f(x) + \mu^2 \sum_x f(x) \\ &= \sum_x x^2 f(x) - 2\mu^2 + \mu^2 \\ &= \sum_x x^2 f(x) - \mu^2 \text{ is the computational formula} \end{aligned}$$

The computational formula is easier to calculate manually.



Exercise: Messages

- ♣ The number of messages sent per hour over a computer network has the following distribution. Find the mean & standard deviation of the number of messages sent per hour.

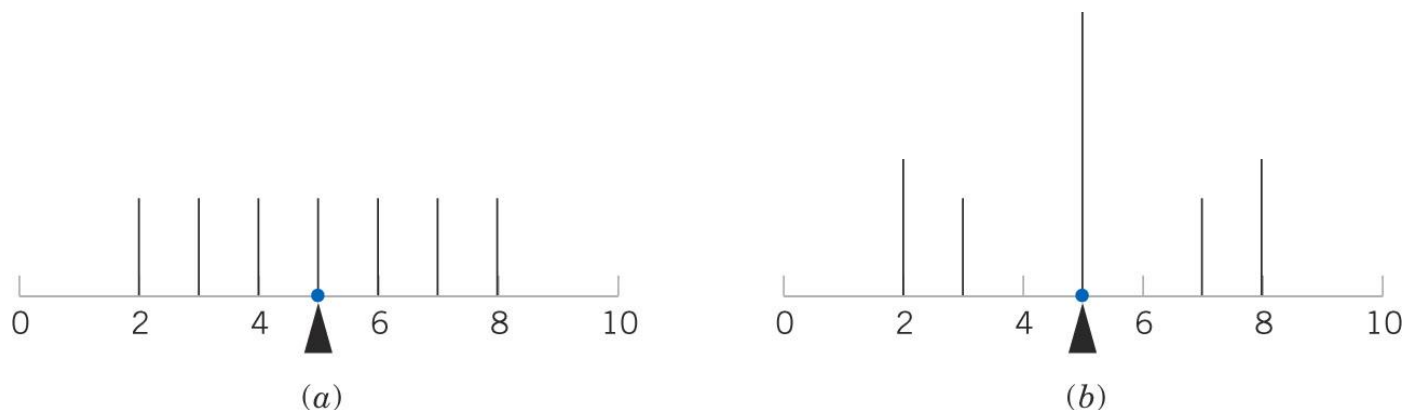
x	$f(x)$
10	0.08
11	0.15
12	0.30
13	0.20
14	0.20
15	0.07
	1.00

$x * f(x)$	$x^2 * f(x)$
0.80	8
1.65	18.15
3.60	43.2
2.60	33.8
2.80	39.2
1.05	15.75



Different Distributions Have Same Measures

- ♣ These measures do not uniquely identify a probability distribution – different distributions could have the same mean & variance.



These probability distributions have the same mean and variance measures, but are very different in shape.



Discrete Uniform Distribution

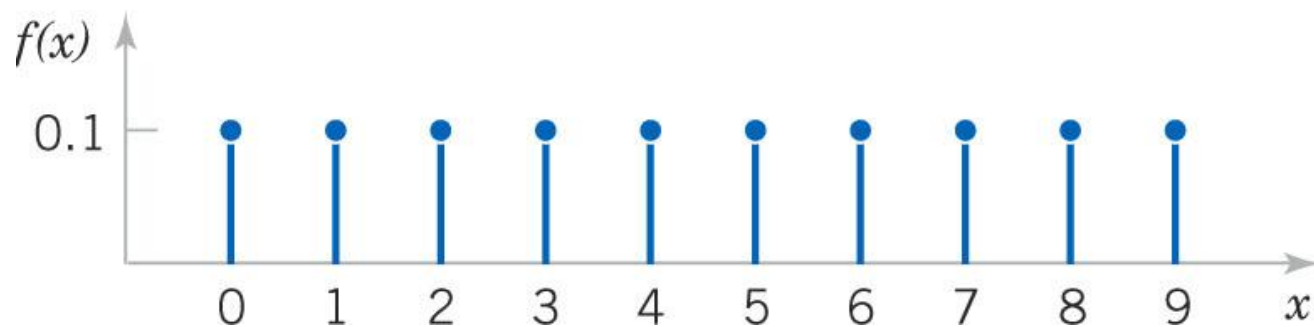
- ♣ Simplest discrete distribution.
- ♣ The random variable X assumes only a finite number of values, each with equal probability.
- ♣ A random variable X has a discrete uniform distribution if each of the n values in its range, say $x_1, x_2, \dots, x_n$, has equal probability.

$$f(x_i) = 1/n$$



Example: Discrete Uniform Random Variable

- ♣ The first digit of a part's serial number is equally likely to be the digits 0 through 9. If one part is selected from a large batch & X is the 1st digit of the serial number, then X has a discrete uniform distribution as shown.



Probability mass function, $f(x) = 1/10$ for $x = 0, 1, 2, \dots, 9$



General Discrete Uniform Distribution

- ♣ Let X be a discrete uniform random variable from a to b for $a < b$. There are $b-(a-1)$ values in the inclusive interval. Therefore:

$$f(x) = 1/(b-a+1)$$

- ♣ Its measures are:

$$\mu = E(x) = \frac{1}{2}^*(a+b)$$

$$\sigma^2 = V(x) = ((b-a+1)^2 - 1) / 12$$

- * Note that the mean is the midpoint of a & b .



Binomial Random Variables

These are binomial experiments having the following characteristics:

1. Fixed number of trials (n).
2. Each trial is termed a **success or failure**. X is the # of successes.
3. The probability of success in each trial is constant (p).
4. The outcomes of successive trials are independent.



Examples of Binomial Random Variables

- ♣ Flip a coin 10 times.
 $X = \#$ heads obtained.
- ♣ A worn tool produces 1% defective parts.
 $X = \#$ defective parts in the next 25 parts produced.
- ♣ A multiple-choice test contains 10 questions, each with 4 choices, and you guess.
 $X = \#$ of correct answers.
- ♣ Of the next 20 births, let $X = \#$ females.



Example: Digital Channel

- ♣ The chance that a bit transmitted through a digital transmission channel is received in error is 0.1.
- ♣ Assume that the transmission trials are independent.
- ♣ Let X = the number of bits in error in the next 4 bits transmitted.
- ♣ Find $P(X=2)$.

♣ **Answer:**

- ◆ Let E denote a bit in error
- ◆ Let O denote an OK bit.
- ◆ Sample space & x listed in table.
- ◆ 6 outcomes where $x = 2$.
 - ♥ Prob of each is $0.1^2 * 0.9^2 = 0.0081$
 - ♥ $\text{Prob}(X=2) = 6 * 0.0081 = 0.0486$

Outcome	x	Outcome	x
OOOO	0	E000	1
OOOE	1	EOOE	2
OOEO	1	EOEO	2
OOEE	2	EOEE	3
OEOO	1	EEOO	2
OEOE	2	EEOE	3
OEE0	2	EEEO	3
OEEE	3	EEEE	4

$$P(X = 2) = C_2^4 (0.1)^2 (0.9)^2$$

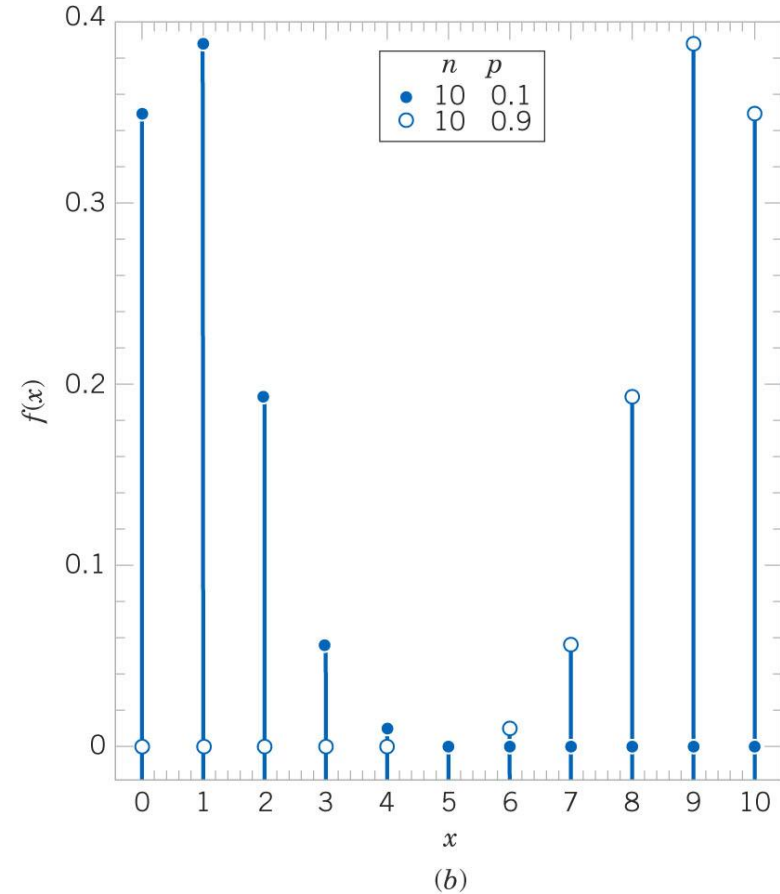
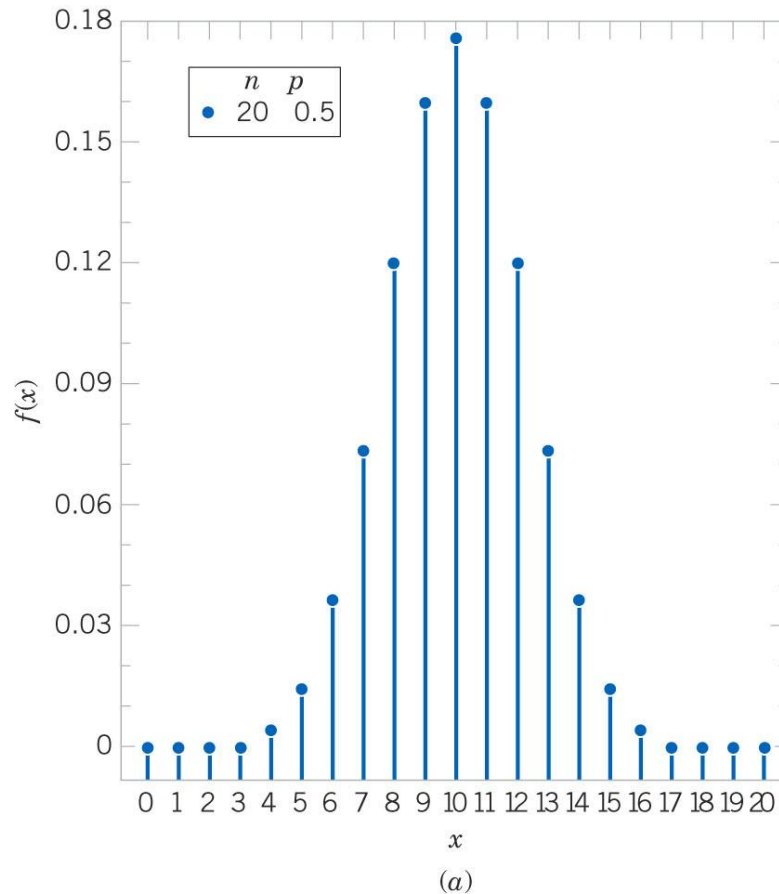


Binomial Distribution Definition

- ♣ The random variable X that equals the number of trials that result in a success is a binomial random variable with parameters $0 < p < 1$ and $n = 0, 1, \dots$
- ♣ The **probability mass function** is:

$$f(x) = C_x^n p^x (1 - p)^{n-x} \text{ for } x = 0, 1, \dots, n$$

Binomial Distribution Shapes



Binomial Distributions for selected values of n and p . Distribution (a) is symmetrical, while distributions (b) are skewed. The skew is right if p is small.



Binomial Mean and Variance

- ♣ If X is a binomial random variable with parameters p and n ,

$$\mu = E(X) = np \quad \text{and} \quad \sigma^2 = V(X) = np(1-p)$$

- ♣ **Example:** For the number of transmitted bit received in error, $n = 4$ and $p = 0.1$.

Find the mean and variance of the binomial random variable.

- ♣ **Answer:**

- ♦ $\mu = E(X) = np = 4 * 0.1 = 0.4$

- ♦ $\sigma^2 = V(X) = np(1-p) = 4 * 0.1 * 0.9 = 0.36$

- ♦ $\sigma = SD(X) = 0.6$



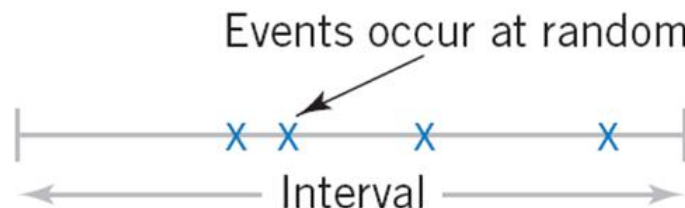
Binomial Random Variables

These are binomial experiments having the following characteristics:

1. Fixed number of trials (n).
2. Each trial is termed a **success or failure**. X is the # of successes.
3. The probability of success in each trial is constant (p).
4. The outcomes of successive trials are independent.

Poisson Process

- ♣ Consider e-mail messages that arrive at a mail server on a computer network.
- ♣ An example of events (message arriving) that occur randomly in an interval (time).
- ♣ The number of events over an interval (e.g. # of messages that arrive in 1 hour) is a **discrete random variable** that is often modelled by a **Poisson distribution**.
- ♣ The length of the interval between events (time between two messages) is often modelled by an **exponential distribution**.



In a Poisson process,
event occur at random
in an interval.



Examples of Poisson Processes

- ♣ In general, the Poisson random variable X is the number of events (counts) per interval.
 1. Particles of contamination per wafer.
 2. Flaws per roll of textile.
 3. Calls at a telephone exchange per hour.
 4. Power outages per year.
 5. Atomic particles emitted from a specimen per second.
 6. Flaws per unit length of copper wire.



Poisson Distribution

- ♣ As the number of trials (n) in a binomial experiment increases to infinity while the binomial mean (np) remains constant, the binomial distribution becomes the Poisson distribution.

- ♣ **Example:**

Let $\lambda = np = E(x)$, so $p = \lambda/n$

$$\begin{aligned} P(X = x) &= \binom{n}{x} p^x (1-p)^{n-x} \\ &= \binom{n}{x} \left(\frac{\lambda}{n}\right)^x \left(1 - \frac{\lambda}{n}\right)^{n-x} \rightarrow \\ &= \frac{e^{-\lambda} \lambda^x}{x!} \end{aligned}$$



Poisson Distribution Definition

- ♣ The random variable X that equals the number of events in a Poisson process is a Poisson random variable with parameter $\lambda > 0$, and the probability mass function is:

$$f(x) = \frac{e^{-\lambda} \lambda^x}{x!} \quad \text{for } x = 0, 1, 2, 3, \dots \infty$$



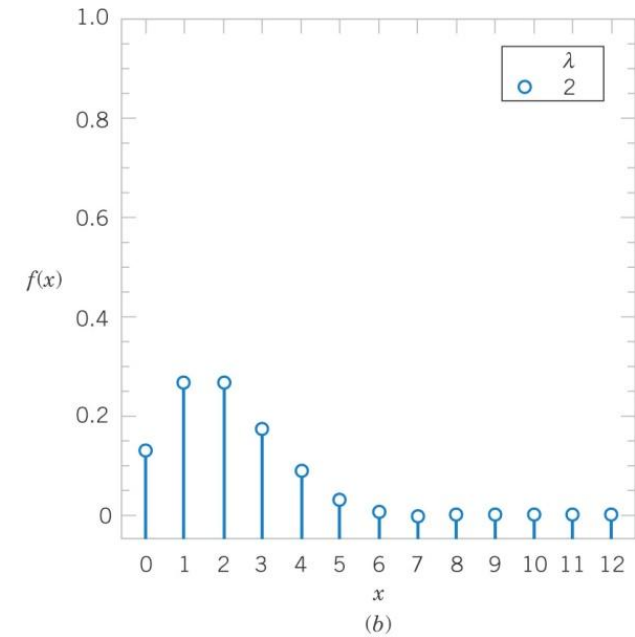
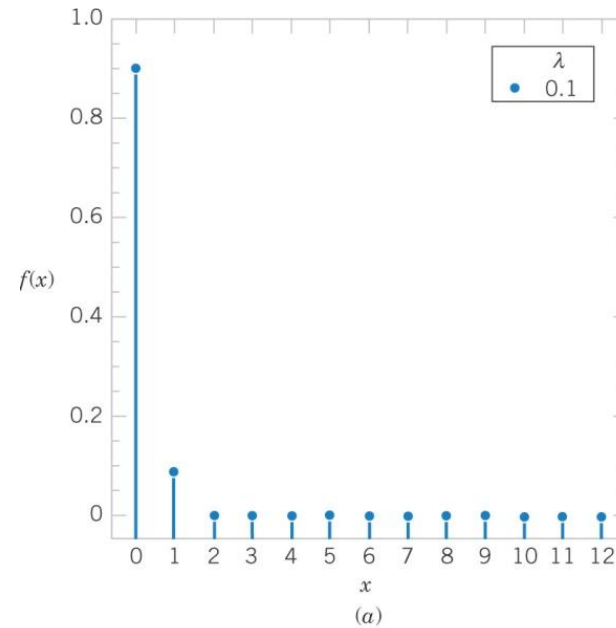
Example: Calculations for Wire Flaws

- ♣ For the case of the thin copper wire, suppose that the number of flaws follows a Poisson distribution of 2.3 flaws per mm.
- ♣ Let X denote the number of flaws in 1 mm of wire.
- ♣ Find the probability of exactly 2 flaws in 1 mm of wire.

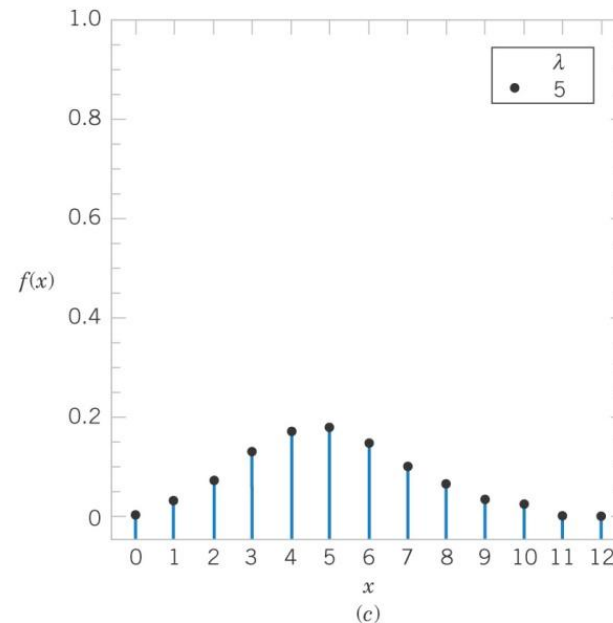
♣ Answer:

$$P(X = 2) = \frac{e^{-2.3} 2.3^2}{2!} = 0.265$$

In Excel	
0.26518	= POISSON(2, 2.3, FALSE)



Poisson Graphs



Poisson distributions for $\lambda = 0.1, 2, 5$.



Poisson Mean & Variance

- ♣ If X is a Poisson random variable with parameter λ , then:

$$\mu = E(X) = \lambda \quad \text{and} \quad \sigma^2 = V(X) = \lambda$$

- ♣ The **mean and variance of the Poisson model are the same**. If the mean and variance of a data set are not about the same, then the Poisson model would not be a good representation of that set.



Poisson Requires Consistent Units

- ♣ It is important to use consistent units in the calculation of Poisson:
 - ◆ Probabilities
 - ◆ Means
 - ◆ Variances

- ♣ Example of unit conversions:
 - ◆ Average # of flaws per mm of wire is 3.4.
 - ◆ Average # of flaws per 10 mm of wire is 34.
 - ◆ Average # of flaws per 20 mm of wire is 68.



Example: Calculations for Wire Flaws

- ♣ For the case of the thin copper wire, suppose that the number of flaws follows a Poisson distribution of 2.3 flaws per mm.
- ♣ Determine the probability of 10 flaws in 5 mm of wire.
- ♣ Now let X denote the number of flaws in 5 mm of wire.
- ♣ The number of flaws follows a Poisson distribution of 2.3 flaws per mm.



Random Variables and Probability Distributions

- ♣ Random Variables and Probability
- ♣ Discrete Random Variables
 - ◆ Binomial Distribution
 - ◆ Poisson Process
- ♣ Continuous Random Variables
- ♣ Normal Approximation to the Binomial and Poisson Distributions
- ♣ More than One Random Variable and Independence
- ♣ Functions of Random Variables



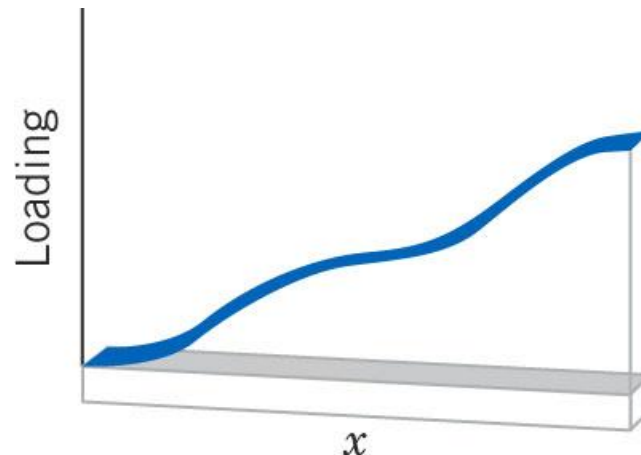


Continuous Random Variables

- ♣ The dimensional length of a manufactured part is subject to small variations in measurement due to vibrations, temperature fluctuations, operator differences, calibration, cutting tool wear, bearing wear, and raw material changes.
- ♣ This length X would be a **continuous random variable** that would occur in an interval (finite or infinite) of real numbers.
- ♣ The number of possible values of X , in that interval, is **uncountably infinite** and limited only by the precision of the measurement instrument.

Continuous Density Functions

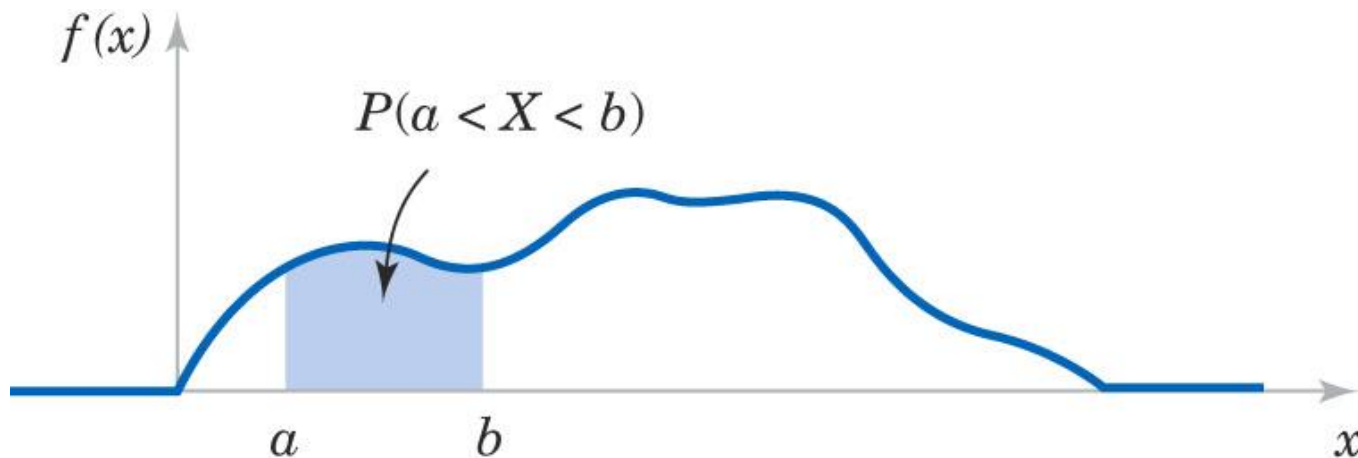
- ♣ **Density functions**, in contrast to mass functions, distribute probability continuously along an interval.
- ♣ The loading on the beam between points a & b is the integral of the function between points a & b .



Density function as a loading on a long, thin beam.
Most of the load occurs at the larger values of x .

Probability Density Function

- ♣ A **probability density function** $f(x)$ describes the probability distribution of a continuous random variable.
- ♣ It is analogous to the beam loading.



Probability is determined from the area under $f(x)$ from a to b .



Probability Density Function

For a continuous random variable X ,
a **probability density function** is a function such that

(1) $f(x) \geq 0$ means that the function is always non-negative.

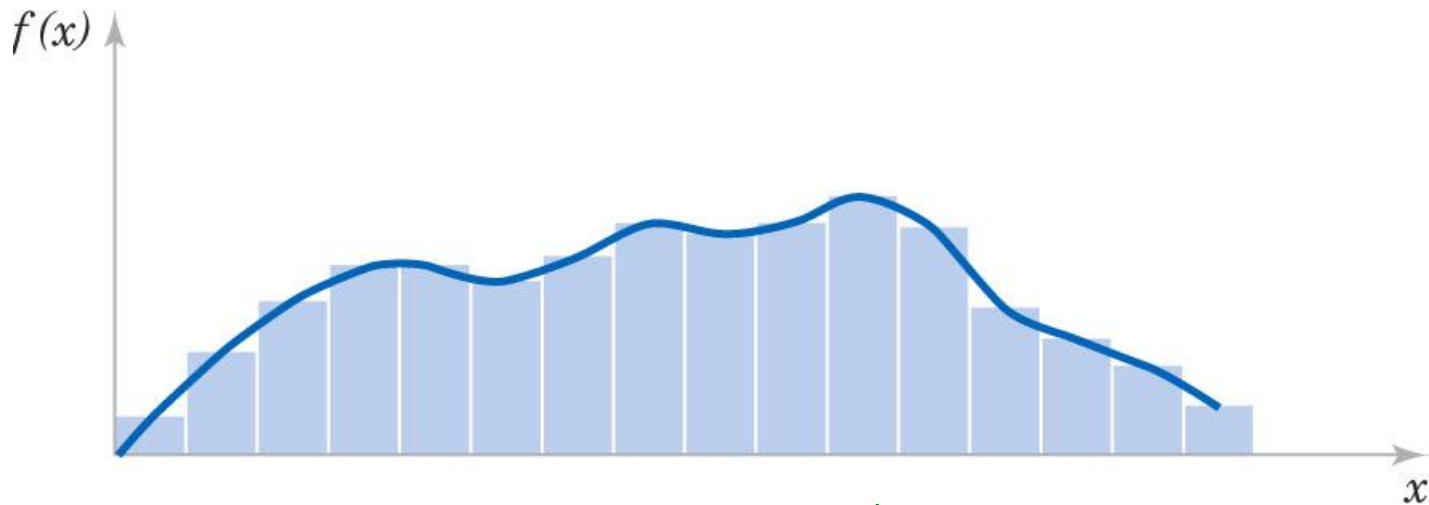
$$(2) \int_{-\infty}^{\infty} f(x)dx = 1$$

$$(3) P(a \leq X \leq b) = \int_a^b f(x)dx = \text{area under } f(x)dx \text{ from } a \text{ to } b$$

(4) $f(x) = 0$ means there is no area exactly at x .

Histograms

- ♣ A **histogram** is graphical display of data showing a series of adjacent rectangles. Each rectangle has a base which represents an interval of data values. The height of the rectangle creates an **area** which represents the relative frequency associated with the values included in the base.
- ♣ A continuous probability distribution $f(x)$ is a model approximating a histogram. A bar has the same area of the integral of those limits.



Histogram approximates a probability density function.



Area of a Point

- ♣ If X is a continuous random variable, for any x_1 and x_2 ,

$$\begin{aligned} P(x_1 \leq X \leq x_2) \\ &= P(x_1 < X \leq x_2) \\ &= P(x_1 \leq X < x_2) \\ &= P(x_1 < X < x_2) \end{aligned}$$

which implies that $P(X = x) = 0$.

- ♣ From another perspective:

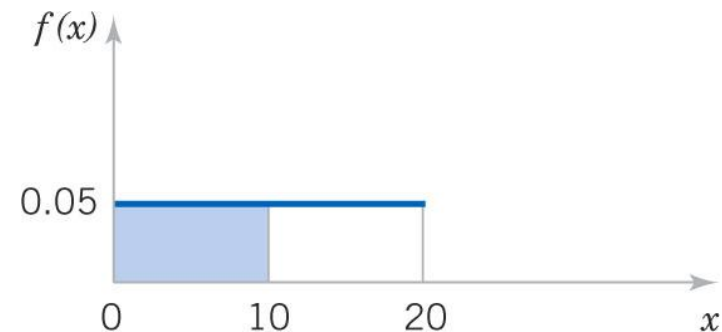
As x_1 approaches x_2 , the area or probability becomes smaller and smaller.

As x_1 becomes x_2 , the area or probability becomes zero.



Example: Electric Current

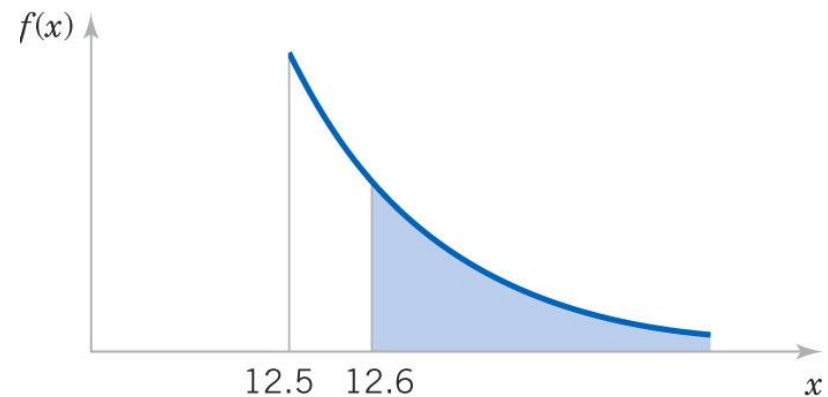
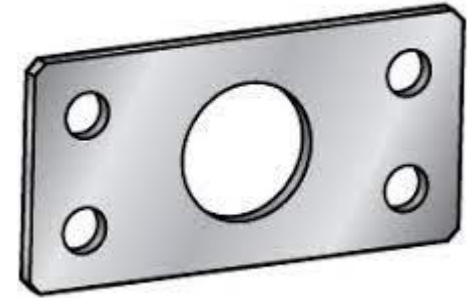
- ♣ Let the continuous random variable X denote the current measured in a thin copper wire in milliamperes (mA).
- ♣ Assume that the range of X is $0 \leq x \leq 20$ and $f(x) = 0.05$. What is the probability that a current is less than 10mA?
- ♣ Answer:



$P(X < 10)$ illustrated.

Example: Hole Diameter

- ♣ Let the continuous random variable X denote the diameter of a hole drilled in a sheet metal component.
- ♣ The target diameter is 12.5 mm. Random disturbances to the process result in larger diameters. Historical data shows that the distribution of X can be modeled by $f(x) = 20e^{-20(x-12.5)}, x \geq 12.5$ mm.
- ♣ If a part with a diameter larger than 12.60 mm is scrapped, what proportion of parts is scrapped?
- ♣ Answer:





Cumulative Distribution Functions

- ♣ The **cumulative distribution function** of a continuous random variable X is,

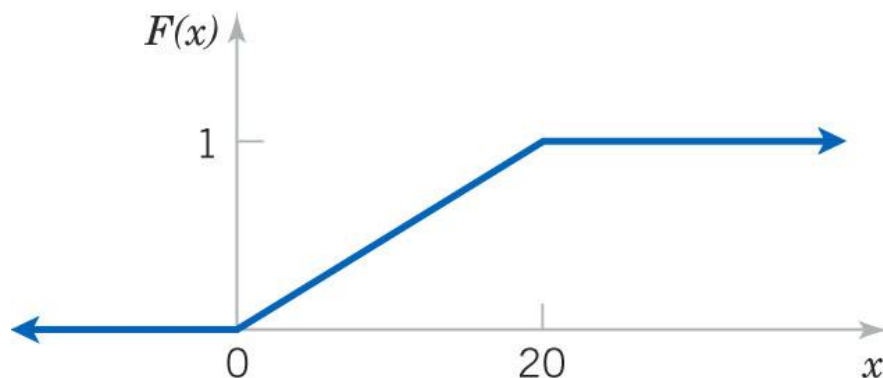
$$F(x) = P(X \leq x) = \int_{-\infty}^x f(u) du \quad \text{for } -\infty < x < \infty$$



Example: Electric Current

- ♣ For the copper wire current measurement, the cumulative distribution function (CDF) consists of three expressions to cover the entire real number line.

	0	$x < 0$
$F(x) =$	$0.05x$	$0 \leq x \leq 20$
	1	$20 < x$



This graph shows the CDF as a continuous function.

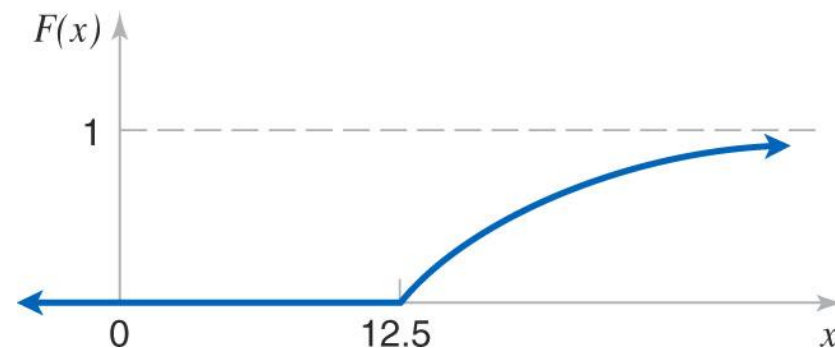


Example: Hole Diameter

- ♣ For the drilling operation, $F(x)$ consists of two expressions.
- ♣ This shows the proper notation.

$$F(x) = 0 \quad \text{for } x < 12.5$$

$$F(x) = \int_{12.5}^x 20e^{-20(u-12.5)} du$$
$$= 1 - e^{-20(x-12.5)} \quad \text{for } x \geq 12.5$$



This graph shows $F(x)$ as a continuous function.



Density vs. Cumulative Functions

- ♣ The probability density function (PDF) is the derivative of the cumulative distribution function (CDF).
- ♣ The cumulative distribution function (CDF) is the integral of the probability density function (PDF).

Given $F(x)$, $f(x) = \frac{dF(x)}{dx}$ as long as the derivative exists.



Example: Reaction Time

- ♣ The time until a chemical reaction is complete (in milliseconds, ms) is approximated by this CDF:

$$F(x) = \begin{cases} 0 & \text{for } x < 0 \\ 1 - e^{-0.01x} & \text{for } 0 \leq x \end{cases}$$

- ♣ What is the PDF?

$$f(x) = \frac{dF(x)}{dx} = \frac{d}{dx} \begin{cases} 0 \\ 1 - e^{-0.01x} \end{cases} = \begin{cases} 0 & \text{for } x < 0 \\ 0.01e^{-0.01x} & \text{for } 0 \leq x \end{cases}$$

- ♣ What proportion of reactions is complete within 200 ms?

$$P(X < 200) = F(200) =$$



Mean & Variance

Suppose X is a continuous random variable with probability density function $f(x)$. The **mean** or **expected value** of X , denoted as μ or $E(X)$, is

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x) dx$$

The **variance** of X , denoted as $V(X)$ or σ^2 , is

$$\sigma^2 = V(X) = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx = \int_{-\infty}^{\infty} x^2 f(x) dx - \mu^2$$

The **standard deviation** of X is $\sigma = \sqrt{\sigma^2}$.



Example: Electric Current

- ♣ For the copper wire current measurement, the PDF is $f(x) = 0.05$ for $0 \leq x \leq 20$.
- ♣ Find the mean and variance.

$$E(X) = \int_0^{20} x \cdot f(x) dx = \frac{0.05x^2}{2} \Big|_0^{20} =$$

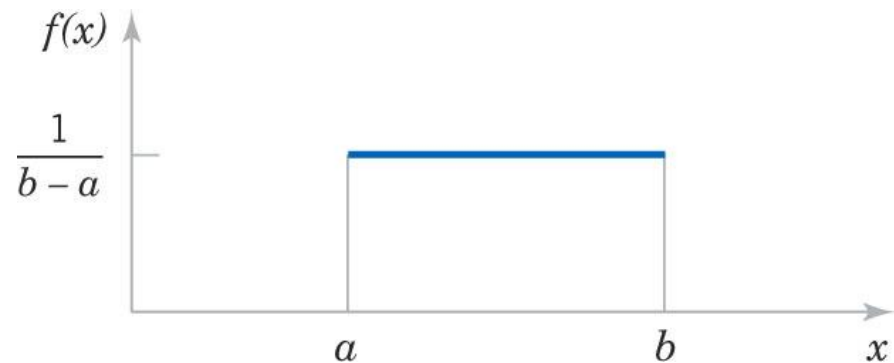
$$V(X) = \int_0^{20} (x-10)^2 f(x) dx = \frac{0.05(x-10)^3}{3} \Big|_0^{20} =$$

Continuous Uniform Distribution

- ♣ This is the simplest continuous distribution and analogous to its discrete counterpart.
- ♣ A continuous random variable X with probability density function

$$f(x) = 1/(b - a)$$

for $a \leq x \leq b$



Continuous uniform PDF

- ♣ Mean & variance are

$$\mu = E(X) = \frac{(a + b)}{2} \quad \text{and} \quad \sigma^2 = V(X) = \frac{(b - a)^2}{12}$$

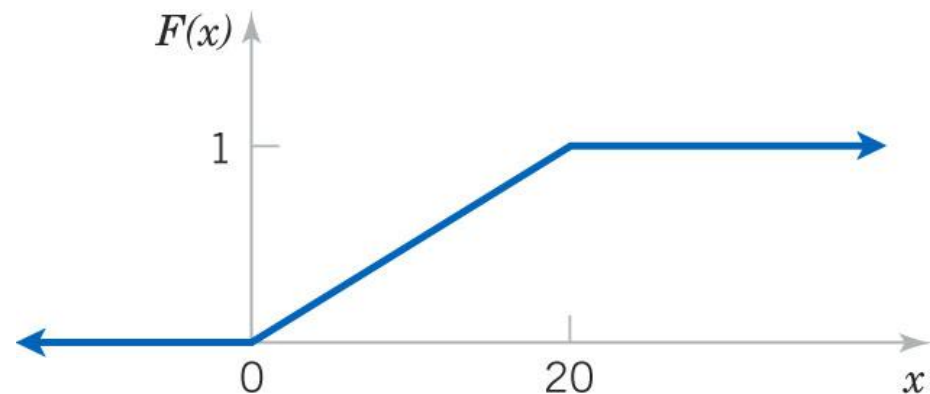


Continuous Uniform CDF

$$F(x) = \int_a^x \frac{1}{(b-a)} du = \frac{x-a}{b-a}$$

The CDF is completely described as

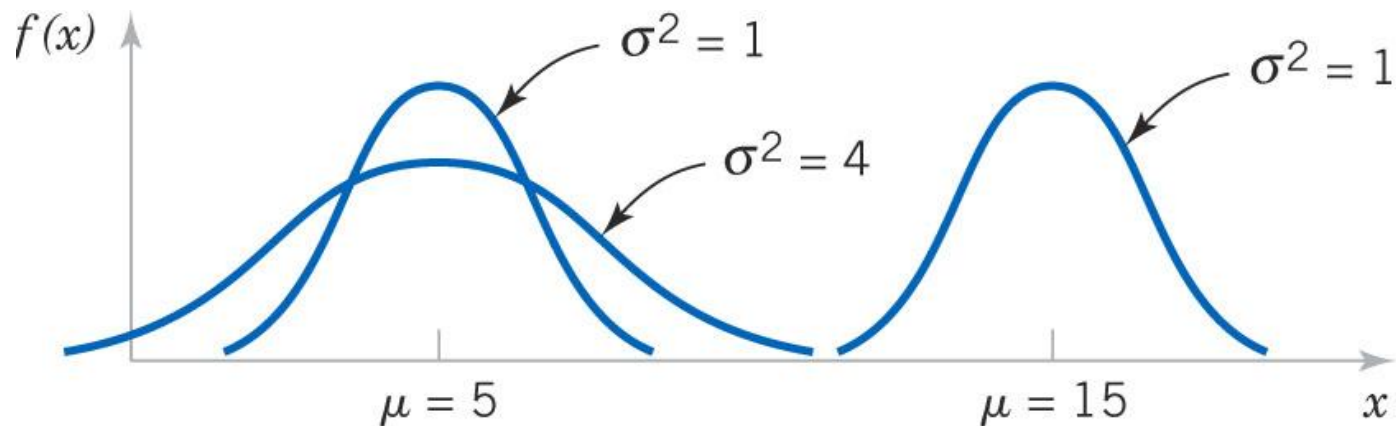
$$F(x) = \begin{cases} 0 & x < a \\ (x-a)/(b-a) & a \leq x < b \\ 1 & b \leq x \end{cases}$$



Graph of the Cumulative Uniform CDF

Normal Distribution

- ♣ The most widely used distribution is the **normal distribution**, also known as the **Gaussian distribution**.
- ♣ Random variation of many physical measurements are normally distributed.
- ♣ The location and spread of the normal are independently determined by mean (μ) and standard deviation (σ).



Normal probability density functions



Normal Probability Density Function

- ♣ A random variable X with probability density function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad -\infty < x < \infty$$

- ♣ is a **normal random variable** with parameters μ , where $-\infty < \mu < \infty$, and $\sigma > 0$.

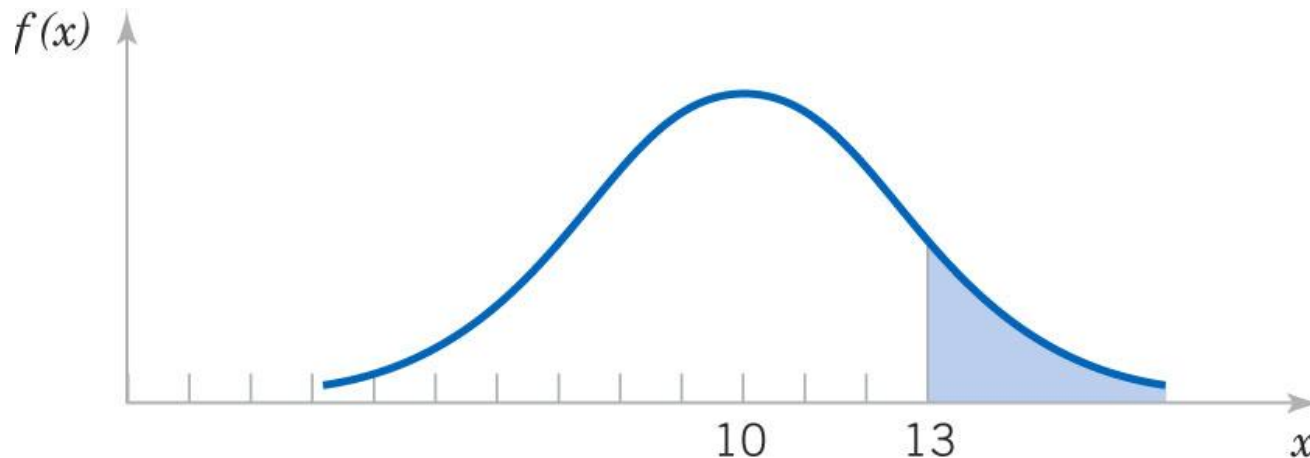
- ♣ Also,

$$E(X) = \mu \quad \text{and} \quad V(X) = \sigma^2$$

- ♣ and the notation $N(\mu, \sigma^2)$ is used to denote the distribution.

Example: Normal Application

- ♣ Assume that the **current measurements** in a strip of wire follows a normal distribution with a **mean of 10 mA** & a **variance of 4 mA²**.
- ♣ Let X denote the current in mA.
- ♣ What is the probability that a measurement exceeds 13 mA?



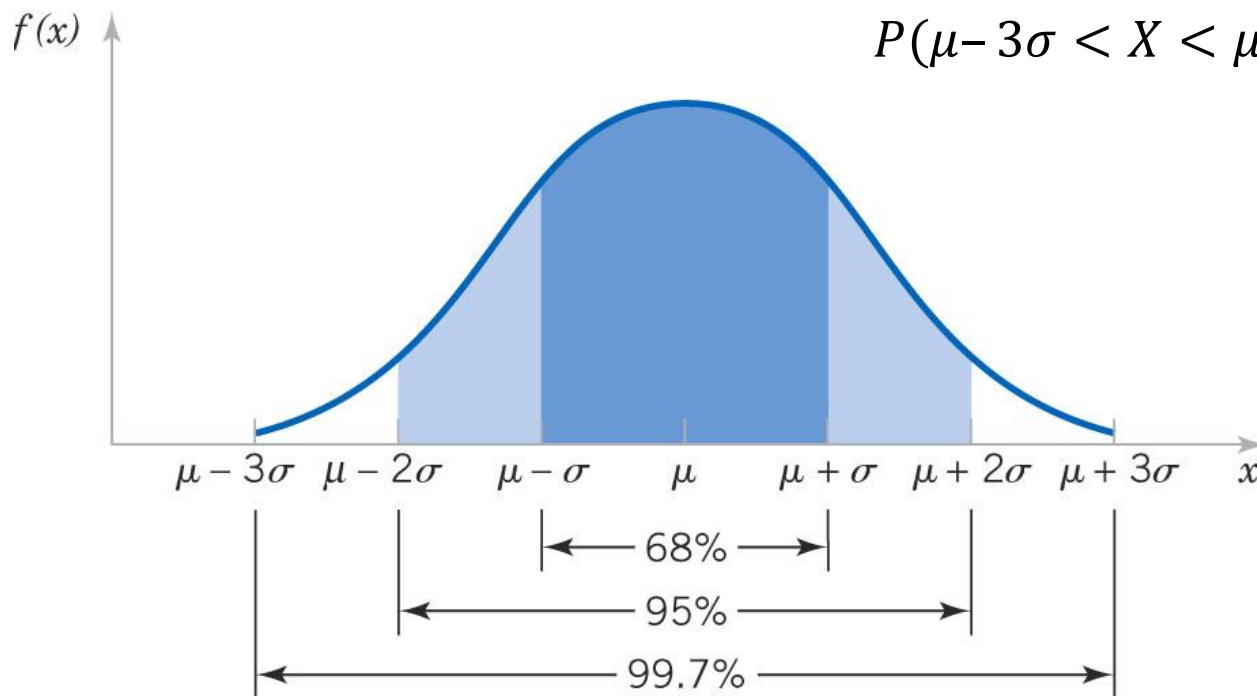
Graphical probability that $X > 13$ for a normal random variable
with $\mu = 10$ and $\sigma^2 = 4$.

Empirical Rule

$$P(\mu - \sigma < X < \mu + \sigma) = 0.6827$$

$$P(\mu - 2\sigma < X < \mu + 2\sigma) = 0.9545$$

$$P(\mu - 3\sigma < X < \mu + 3\sigma) = 0.9973$$



Probabilities associated with a normal distribution – well worth remembering to quickly estimate probabilities.



Standard Normal Distribution

- ♣ A normal random variable with

$$\mu = 0 \text{ and } \sigma^2 = 1$$

is called a **standard normal random variable** and is denoted as Z .

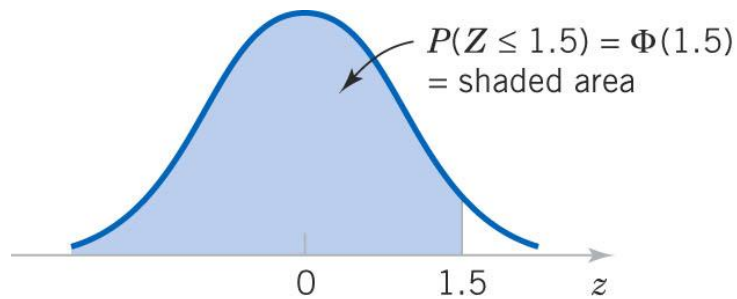
- ♣ The cumulative distribution function of a standard normal random variable is denoted as:

$$\Phi(z) = P(Z \leq z) = F(z)$$



Example: Standard Normal Distribution

- ♣ Assume Z is a standard normal random variable.
- ♣ Find $P(Z \leq 1.50)$.



z	0.00	0.01	0.02	0.03
0	0.50000	0.50399	0.50398	0.51197
$\vdots$		$\vdots$		
1.5	0.93319	0.93448	0.93574	0.93699

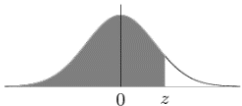
Standard normal PDF

- ♣ Find $P(Z \leq 1.53)$.
- ♣ Find $P(Z \leq 0.02)$.



CUMULATIVE NORMAL DISTRIBUTION TABLE

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-s^2/2} ds$$



Cumulative Distribution Table

[illegible]



Standardizing

Suppose X is a normal random variable with mean μ and variance σ^2 .

$$\text{Then, } P(X \leq x) = P\left(\frac{X - \mu}{\sigma} \leq \frac{x - \mu}{\sigma}\right) = P(Z \leq z)$$

where Z is a **standard normal random variable**, and

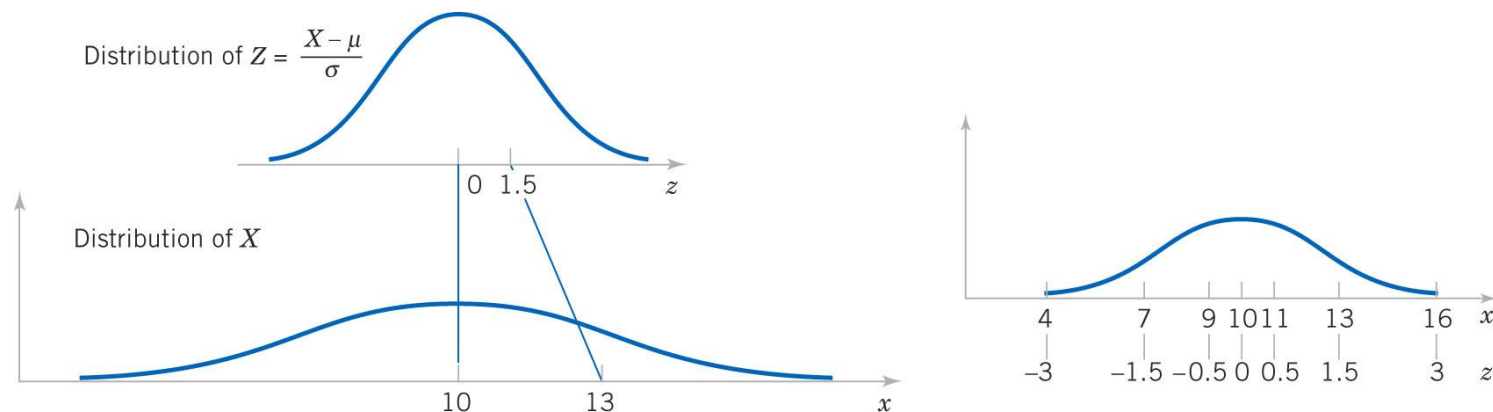
$z = \frac{(x - \mu)}{\sigma}$ is the z-value obtained by **standardizing** X .

Example: Normally Distributed Current-1

- ♣ From a previous example with $\mu = 10$ and $\sigma = 2$ mA, what is the probability that the current measurement is between 9 and 11 mA?

♣ Answer:

$$\begin{aligned}
 P(9 < X < 11) &= P\left(\frac{9 - 10}{2} < \frac{X - 10}{2} < \frac{11 - 10}{2}\right) \\
 &= P(-0.5 < Z < 0.5) \\
 &= P(Z < 0.5) - P(Z < -0.5)
 \end{aligned}$$



Standardizing a normal random variable.



Example: Normally Distributed Current-2

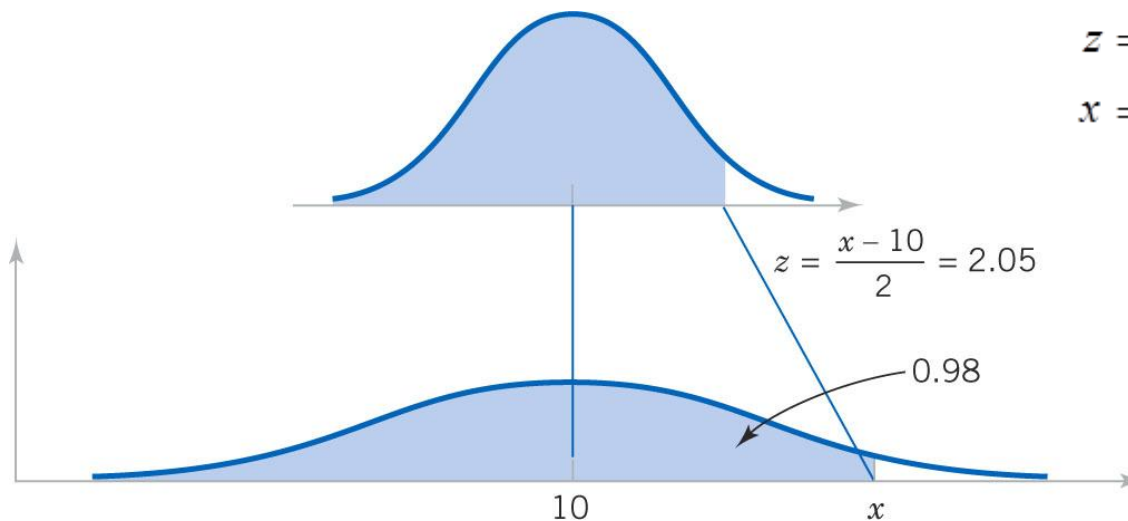
- ♣ Determine the value for which the probability that a current measurement is below this value is 0.98.

♣ Answer:

$$\begin{aligned} P(X < x) &= P\left(\frac{X - 10}{2} < \frac{x - 10}{2}\right) \\ &= P\left(Z < \frac{x - 10}{2}\right) = 0.98 \end{aligned}$$

$z =$

$x =$



Determining the value of x to meet a specified probability.



Example: Signal Detection

- ♣ Assume that in the detection of a digital signal, the **background noise** follows a normal distribution with $\mu = 0$ volt and $\sigma = 0.45$ volt.
- ♣ The system assumes a **signal 1 has been transmitted when the voltage exceeds 0.9**. What is the probability of detecting a digital 1 when none was sent?

- ♣ Let the random variable N denote the voltage of noise.

$$P(N > 0.9) = P\left(\frac{N - 0}{0.45} > \frac{0.9}{0.45}\right) = P(Z > 2)$$

=

- ♣ This probability can be described as the probability of a false detection.



Example : Signal Detection-2

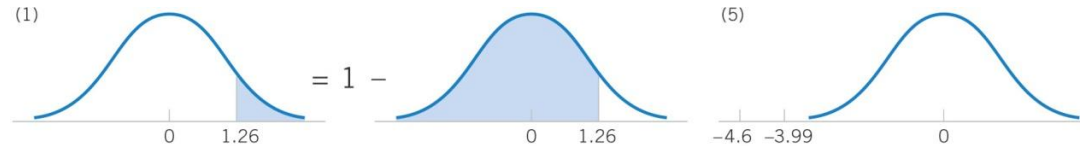
- ♣ Suppose that when a digital 1 signal is transmitted, the mean of the noise distribution shifts to 1.8 volts. What is the probability that a digital 1 is not detected?
- ♣ Let S denote the voltage when a digital 1 is transmitted.

$$\begin{aligned} P(S < 0.9) &= P\left(\frac{S - 1.8}{0.45} < \frac{0.9 - 1.8}{0.45}\right) \\ &= \end{aligned}$$

- ♣ This probability can be interpreted as the probability of a missed signal.

Standard Normal Exercises

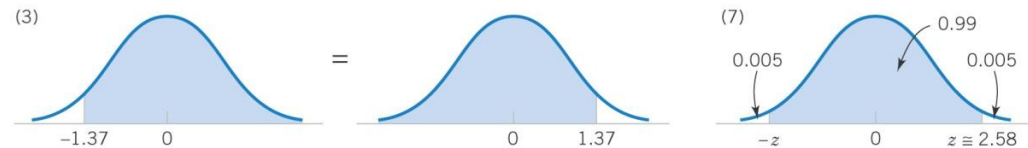
1. $P(Z > 1.26) =$



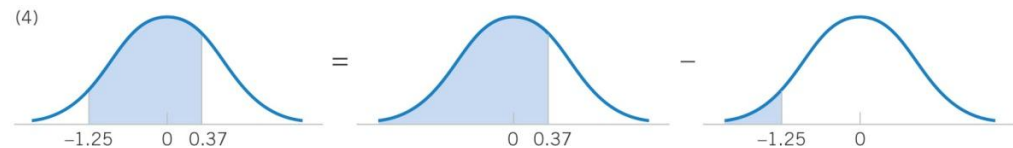
2. $P(Z < -0.86) =$



3. $P(Z > -1.37) =$



4. $P(-1.25 < Z < 0.37) =$



5. $P(Z \leq -4.6) \approx$

6. Find z for $P(Z \leq z) = 0.05$, $z =$

7. Find z for $P(-z < Z < z) = 0.99$, $z =$



Random Variables and Probability Distributions

- ♣ Random Variables and Probability
- ♣ Discrete Random Variables
 - ◆ Binomial Distribution
 - ◆ Poisson Process
- ♣ Continuous Random Variables
- ♣ Normal Approximation to the Binomial and Poisson Distributions
- ♣ More than One Random Variable and Independence
- ♣ Functions of Random Variables





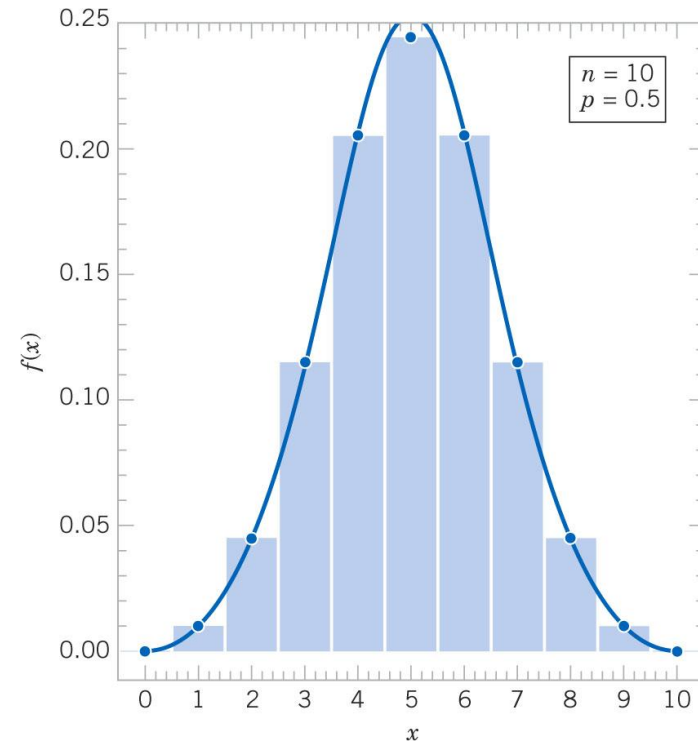
Normal Approximations

- ♣ The binomial and Poisson distributions become more bell-shaped and symmetric as their means increase.
- ♣ For manual calculations, the normal approximation is practical – exact probabilities of the binomial and Poisson, with large means, require technology (Minitab, Excel).
- ♣ The normal is a good approximation for the:
 - ◆ Binomial if $np > 5$ and $n(1-p) > 5$.
 - ◆ Poisson if $\lambda > 5$.



Normal Approximation to the Binomial

- ♣ Suppose we have a **binomial distribution** with $n = 10$ and $p = 0.5$. Its mean and standard deviation are 5.0 and 1.58 respectively.
- ♣ Draw the normal distribution over the binomial distribution.
- ♣ The areas of the normal approximate the areas of the bars of the binomial with a **continuity correction**.



Overlaying the normal distribution upon a binomial with matched parameters.



Example

- ♣ In a digital comm channel, assume that **the number of bits received in error** can be modeled by a binomial random variable.
- ♣ The probability that a bit is received in error is 10^{-5} . If 16 million bits are transmitted, what is the probability that 150 or fewer errors occur?
- ♣ Let X denote the number of errors.
- ♣ **Answer:**

$$P(X \leq 150) = \sum_{x=0}^{150} C_x^{16000000} (10^{-5})^x (1-10^{-5})^{16000000-x}$$

- ♣ Can only be evaluated with technology. Manually, we must use the normal approximation to the binomial.



Normal Approximation Method

If X is a binomial random variable with parameters n and p ,

$$Z = \frac{X - np}{\sqrt{np(1-p)}}$$

is approximately a standard normal random variable. To approximate a binomial probability with a normal distribution, a **continuity correction** is applied as follows:

$$P(X \leq x) = P(X \leq x + 0.5) = P\left(Z \leq \frac{x + 0.5 - np}{\sqrt{np(1-p)}}\right)$$

and

$$P(X < x) = P(X \leq x - 0.5) = P\left(Z \leq \frac{x - 0.5 - np}{\sqrt{np(1-p)}}\right)$$

The approximation is good for $np > 5$ and $n(1-p) > 5$.



Example: Applying the Approximation

- ♣ The digital comm problem in the previous example is solved using the normal approximation to the binomial as follows:

$$\begin{aligned}P(X \leq 150) &= P(X \leq 150.5) \\&= P\left(\frac{X - 160}{\sqrt{160(1 - 10^{-5})}} \leq \frac{150.5 - 160}{\sqrt{160(1 - 10^{-5})}}\right) \\&= P\left(Z \leq \frac{-9.5}{12.6491}\right) \\&= P(Z \leq -0.75104) \\&= \end{aligned}$$



Example: Normal Approximation-1

- ♣ Again consider the transmission of bits.
- ♣ To judge how well the normal approximation works, assume $n = 50$ bits are transmitted and the probability of an error is $p = 0.1$.
- ♣ The exact and approximated probabilities are:

$$P(X \leq 2) = C_0^{50} 0.9^{50} + C_1^{50} 0.1(0.9^{49}) + C_2^{50} 0.1^2(0.9^{48}) = 0.112$$

$$\begin{aligned} P(X \leq 2) &= P\left(\frac{X - 5}{\sqrt{50(0.1)(0.9)}} < \frac{2.5 - 5}{\sqrt{50(0.1)(0.9)}}\right) \\ &= P(Z < -1.18) = \end{aligned}$$



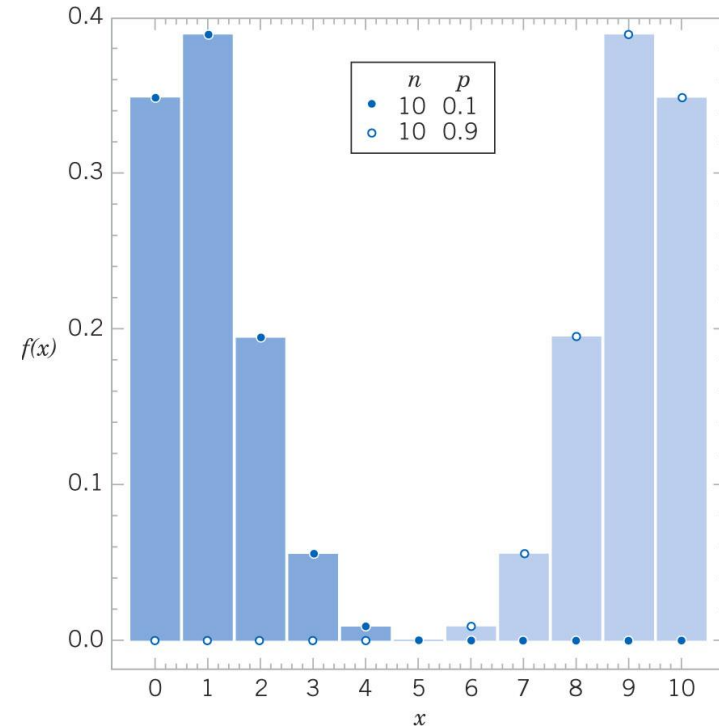
Example: Normal Approximation-2

$$\begin{aligned}P(X > 8) &= P(X \geq 9) \approx P(X > 8.5) \\&= P\left(Z > \frac{8.5 - 5}{2.12}\right) = P(Z > 1.65) =\end{aligned}$$

$$\begin{aligned}P(X = 5) &\approx P(4.5 < X < 5.5) \\&= P\left(\frac{4.5 - 5}{2.12} < Z < \frac{5.5 - 5}{2.12}\right) \\&= P(-0.24 < Z < 0.24) \\&= P(Z < 0.24) - P(Z < -0.24) =\end{aligned}$$

Reason for the Approximation Limits

- ♣ The $np > 5$ and $n(1-p) > 5$ approximation rule is needed to **keep the tails of the normal distribution** from getting out-of-bounds.
- ♣ As the binomial mean approaches **the endpoints of the range of x** , the standard deviation must be small enough to prevent overrun.
- ♣ Figure shows the asymmetric shape of the binomial when the approximation rule is not met.



Binomial distribution is not symmetric as p gets near 0 or 1.



Normal Approximation to the Poisson

- ♣ If X is a Poisson random variable with $E(X) = \lambda$ and $V(X) = \lambda$,

$$Z = \frac{X - \lambda}{\sqrt{\lambda}}$$

is approximately a standard normal random variable. The **same continuity correction** used for the binomial distribution can also be applied. The approximation is good for

$$\lambda \geq 5$$

Example: Normal Approximation to Poisson

- ♣ Assume that the number of asbestos particles in a square meter of dust on a surface follows a **Poisson distribution with a mean of 1000**.
- ♣ If a square meter of dust is analyzed, what is the probability that **950 or fewer particles are found**?



$$P(X \leq 950) = \sum_{x=0}^{950} \frac{e^{-1000} 1000^x}{x!} \quad \dots \text{too hard manually!}$$

≈

=

=



Exercise

- ♣ Each sample of water has a 10% chance of containing a particular organic pollutant. Assume that the samples are independent with regard to the presence of the pollutant.
- ♣ **Approximate** the probability that, in the next 200 samples, there are 20 to 25 samples contain the pollutant.



Exercise

- ♣ For the case of the thin copper wire, suppose that the number of flaws follows a Poisson distribution of 23 flaws per cm.
- ♣ Let X denote the number of flaws in 1 mm of wire.
- ♣ **Approximate** the probability of less than 2 flaws in 1 mm of wire.



Random Variables and Probability Distributions

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- ♣ Functions of Random Variables





Concept of Joint Probabilities

- ♣ Some random variables are not independent of each other, i.e., they tend to be related.
 - ◆ Urban atmospheric ozone and airborne particulate matter tend to vary together.
 - ◆ Vehicle speeds and fuel consumption rates tend to vary inversely.
 - ◆ The length (X) of a injection-molded part might not be independent of the width (Y). Individual parts will vary due to random variation in materials and pressure.
- ♣ A joint probability distribution will describe the behavior of several random variables, say, X and Y . The graph of the distribution is 3-dimensional: x , y , and $f(x, y)$.



Example: Signal Bars

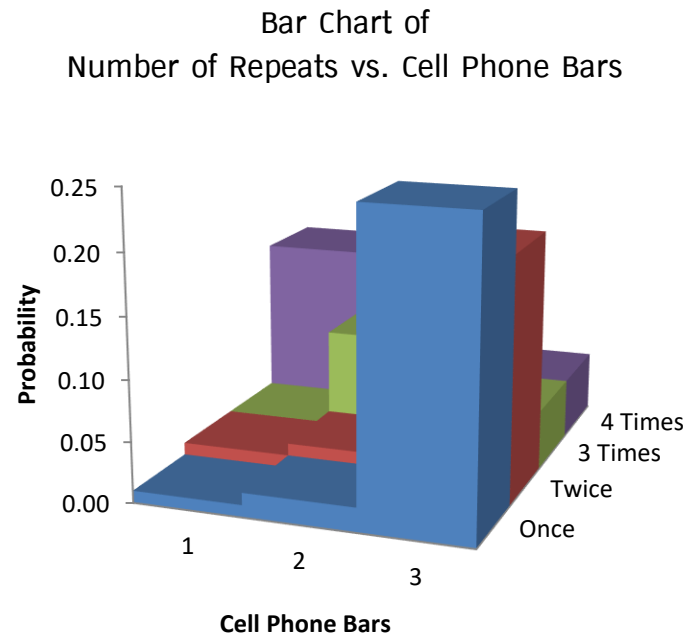
- ♣ You use your cell phone to check your airline reservation. The airline system requires that you speak the name of your departure city to the voice recognition system.
- Let Y denote the number of times that you have to state your departure city.
 - Let X denote the number of bars of signal strength on you cell phone.

y = number of times city name is stated	x = number of bars of signal strength		
	1	2	3
1	0.01	0.02	0.25
2	0.02	0.03	0.20
3	0.02	0.10	0.05
4	0.15	0.10	0.05

Joint probability distribution of X and Y .

The table cells are the probabilities.

Observe that more bars relate to less repeating.





Joint Probability Mass Function Defined

♣ The **joint probability mass function** of the discrete random variables X and Y , denoted as $f_{XY}(x, y)$, satisfies:

♦ $f_{XY}(x, y) \geq 0$

All probabilities are non-negative

♦ $\sum_x \sum_y f_{XY}(x, y) = 1$

The sum of all probabilities is 1

♦ $f_{XY}(x, y) = P(X = x, Y = y)$



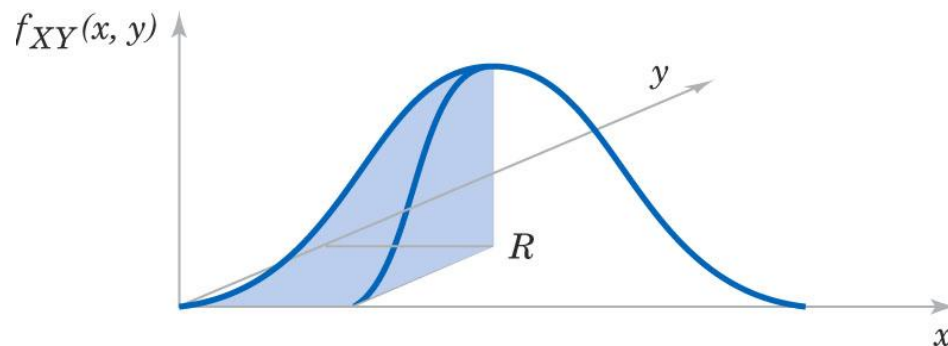
Joint Probability Density Function Defined

♣ The **joint probability density function** for the continuous random variables X and Y , denoted as $f_{XY}(x, y)$, satisfies the following properties:

(1) $f_{XY}(x, y) \geq 0$ for all x, y

(2)
$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = 1$$

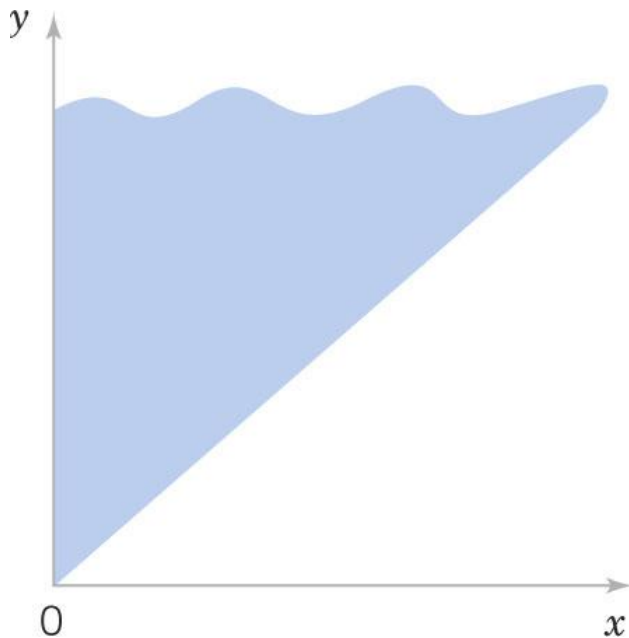
(3)
$$P((X, Y) \in R) = \iint_R f_{XY}(x, y) dx dy$$



Joint probability density function for the random variables X and Y . Probability that (X, Y) is in the region R is determined by the **volume** of $f_{XY}(x, y)$ over the region R .

Example: Server Access Time-1

- ♣ Let the random variable X denote the time (msec's) until a computer server connects to your machine. Let Y denote the time until the server authorizes you as a valid user. X and Y measure the wait from a common starting point ($x < y$). The range of x and y are shown here.



The joint probability density function of X and Y is nonzero over the shaded region where $x < y$.



Example: Server Access Time-2

- ♣ The joint probability density function is:

$$f_{XY}(x, y) = ke^{-0.001x - 0.002y} \text{ for } 0 < x < y < \infty \text{ and } k = 6 \cdot 10^{-6}$$

- ♣ We verify that it integrates to 1 as follows:

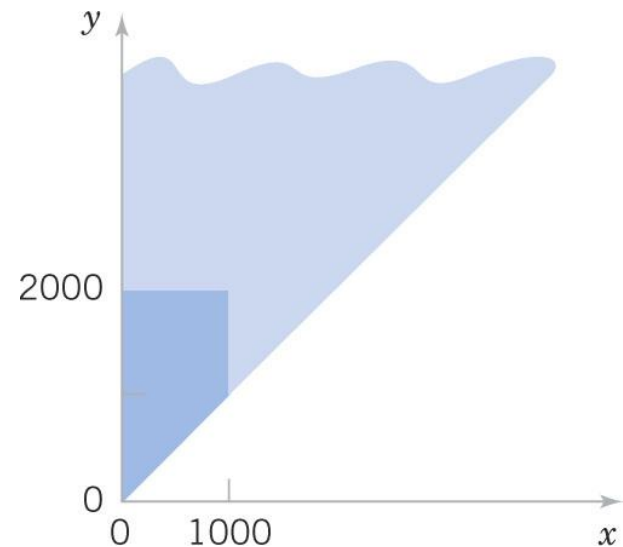
$$\begin{aligned} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy &= \int_0^{\infty} \left(\int_x^{\infty} ke^{-0.001x - 0.002y} dy \right) dx = k \int_0^{\infty} \left(\int_x^{\infty} e^{-0.002y} dy \right) e^{-0.001x} dx \\ &= k \int_0^{\infty} \left(\frac{e^{-0.002x}}{0.002} \right) e^{-0.001x} dx = 0.003 \int_0^{\infty} e^{-0.003x} dx \\ &= 0.003 \left(\frac{1}{0.003} \right) = 1 \end{aligned}$$



Example: Server Access Time-3

Now calculate a probability:

$$\begin{aligned} P(X \leq 1000, Y \leq 2000) &= \int_{-\infty}^{1000} \int_x^{2000} f_{XY}(x, y) dy dx \\ &= k \int_0^{1000} \left(\int_x^{2000} e^{-0.002y} dy \right) e^{-0.001x} dx \\ &= k \int_0^{1000} \left(\frac{e^{-0.002x} - e^{-4}}{0.002} \right) e^{-0.001x} dx \\ &= 0.003 \int_0^{1000} e^{-0.003x} - e^{-4} e^{-0.001x} dx \\ &= 0.003 \left[\left(\frac{1 - e^{-3}}{0.003} \right) - e^{-4} \left(\frac{1 - e^{-1}}{0.001} \right) \right] \\ &= \end{aligned}$$



Region of integration for the probability that $X < 1000$ and $Y < 2000$ is darkly shaded.



Marginal Probability Distributions (discrete)

- ♣ For a **discrete** joint PDF, there are **marginal distributions** for each random variable, formed by summing the joint PMF over the other variable.

$$f_X(x) = \sum_y f(xy)$$

$$f_Y(y) = \sum_x f(xy)$$

y = number of times city name is stated	x = number of bars of signal strength			$f(y) =$
	1	2	3	
1	0.01	0.02	0.25	0.28
2	0.02	0.03	0.20	0.25
3	0.02	0.10	0.05	0.17
4	0.15	0.10	0.05	0.30
$f(x) =$	0.20	0.25	0.55	1.00

From the prior example, the joint PMF is shown in green while the two marginal PMFs are shown in blue.



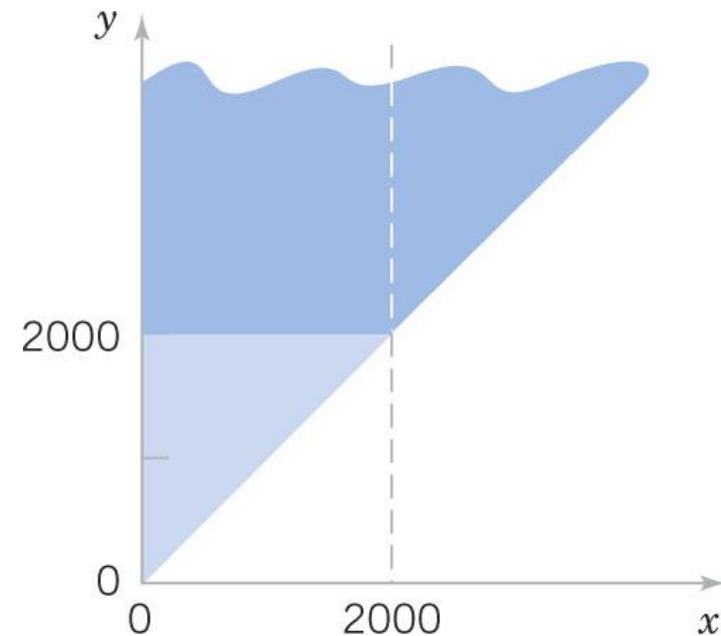
Marginal Probability Distributions (continuous)

- ♣ Rather than summing a discrete joint PMF, we integrate a continuous joint PDF.
- ♣ The marginal PDFs are used to make probability statements about one variable.
- ♣ If the **joint probability density function** of random variables X and Y is $f_{XY}(x,y)$, the marginal probability density functions of X and Y are:

$$f_X(x) = \int_y f_{XY}(x, y) dy$$
$$f_Y(y) = \int_x f_{XY}(x, y) dx$$

Example: Server Access Time-1

- ♣ For the random variables times, find **the probability that Y exceeds 2000**.
- ♣ Integrate the joint PDF directly using the picture to determine the limits.



$$P(Y \geq 2000) = \int_0^{2000} \left(\int_{2000}^{\infty} f_{XY}(x, y) dy \right) dx + \int_{2000}^{\infty} \left(\int_x^{\infty} f_{XY}(x, y) dy \right) dx$$

Dark region = left dark region + right dark region



Example: Server Access Time-2

- ♣ Alternatively, find the marginal PDF and then integrate that to find the desired probability.

$$\begin{aligned}f_Y(y) &= \int_0^y k e^{-0.001x-0.002y} dx \\&= k e^{-0.002y} \int_0^y e^{-0.001x} dx \\&= k e^{-0.002y} \left(\frac{e^{-0.001x}}{-0.001} \right) \Big|_0^y \\&= k e^{-0.002y} \left(\frac{1 - e^{-0.001y}}{0.001} \right) \\&= 6 \cdot 10^{-3} e^{-0.002y} (1 - e^{-0.001y}) \text{ for } y > 0\end{aligned}$$

$$\begin{aligned}P(Y \geq 2000) &= \int_{2000}^{\infty} f_Y(y) dy \\&= 6 \cdot 10^{-3} \int_{2000}^{\infty} e^{-0.002y} (1 - e^{-0.001y}) dy \\&= 6 \cdot 10^{-3} \left[\left(\frac{e^{-0.002y}}{-0.002} \right) \Big|_{2000}^{\infty} - \left(\frac{e^{-0.003y}}{-0.003} \right) \Big|_{2000}^{\infty} \right] \\&= 6 \cdot 10^{-3} \left[\frac{e^{-4}}{0.002} - \frac{e^{-6}}{0.003} \right] = 0.05\end{aligned}$$



Mean & Variance of a Marginal Distribution

- ♣ Means $E(X)$ and $E(Y)$ are calculated from the discrete and continuous marginal distributions.

Discrete

$$E(X) = \sum_R x \cdot f_X(x)$$

$$E(Y) = \sum_R y \cdot f_Y(y)$$

$$V(X) = \sum_R x^2 \cdot f_X(x) - \mu_X^2$$

$$V(Y) = \sum_R y^2 \cdot f_Y(y) - \mu_Y^2$$

Continuous

$$= \int_R x \cdot f_X(x) dx = \mu_X$$

$$= \int_R y \cdot f_Y(y) dy = \mu_Y$$

$$= \int_R x^2 \cdot f_X(x) dx - \mu_X^2$$

$$= \int_R y^2 \cdot f_Y(y) dy - \mu_Y^2$$



Mean & Variance

y = number of times city name is stated	x = number of bars of signal strength					
	1	2	3	$f(y) =$	$y * f(y) =$	$y^2 * f(y) =$
1	0.01	0.02	0.25	0.28	0.28	0.28
2	0.02	0.03	0.20	0.25	0.50	1.00
3	0.02	0.10	0.05	0.17	0.51	1.53
4	0.15	0.10	0.05	0.30	1.20	4.80
$f(x) =$	0.20	0.25	0.55	1.00	2.49	7.61
$x * f(x) =$	0.20	0.50	1.65	2.35		
$x^2 * f(x) =$	0.20	1.00	4.95	6.15		

$$E(X) =$$

$$V(X) =$$

$$E(Y) =$$

$$V(Y) =$$



Conditional Probability Density Function Defined

Given continuous random variables X and Y with joint probability density function $f_{XY}(x, y)$, the conditional probability density function of Y given $X=x$ is

$$f_{Y|x}(y) = \frac{f_{XY}(x, y)}{f_X(x)} \quad \text{for } f_X(x) > 0$$

which satisfies the following properties:

(1) $f_{Y|x}(y) \geq 0$

(2) $\int f_{Y|x}(y) dy = 1$

(3) $P(Y \in B | X = x) = \int_B f_{Y|x}(y) dy$ for any set B in the range of Y



Conditional Probability Distributions

$$\text{Recall that } P(B|A) = \frac{P(A \cap B)}{P(A)}$$

From Signal Bar Example

$$P(Y=1|X=3) =$$

$$P(Y=2|X=3) =$$

$$P(Y=3|X=3) =$$

$$P(Y=4|X=3) =$$

$$\text{Sum} = 1.001$$

y = number of times city name is stated	x = number of bars of signal strength			f(y) =
	1	2	3	
1	0.01	0.02	0.25	0.28
2	0.02	0.03	0.20	0.25
3	0.02	0.10	0.05	0.17
4	0.15	0.10	0.05	0.30
f(x) =	0.20	0.25	0.55	1.00

Note that there are 12 probabilities conditional on X , and 12 more probabilities conditional upon Y .



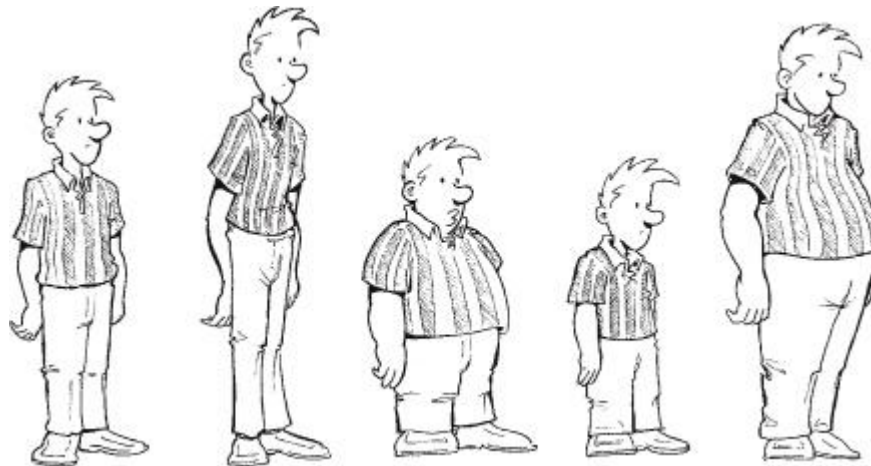
Example: Conditional Discrete PMFs

- ♣ Conditional discrete PMFs can be shown as tables.

y = number of times city name is stated	x = number of bars of signal strength			$f(y) =$	$f(x y)$ for $y =$			Sum of $f(x y) =$
	1	2	3		1	2	3	
1	0.01	0.02	0.25	0.28	A?	0.071	0.893	1.000
2	0.02	0.03	0.20	0.25	0.080	0.120	0.800	1.000
3	0.02	0.10	0.05	0.17	0.118	0.588	B?	1.000
4	0.15	0.10	0.05	0.30	0.500	0.333	0.167	1.000
$f(x) =$	0.20	0.25	0.55					
1	0.050	0.080	0.455					
2	0.100	C?	0.364					
3	0.100	0.400	0.091					
4	D?	0.400	0.091					
Sum of $f(y x) =$	1.000	1.000	1.000					

Joint Random Variable Independence

- ♣ **Random variable independence** means that knowledge of the values of X does not change any of the probabilities associated with the values of Y .
 X and Y are independently.
- ♣ Dependence implies that the values of X are influenced by the values of Y .
- ♣ Do you think that a person's height and weight are independent?





Properties of Independence

- ♣ For random variables X and Y , if any one of the following properties is true, the others are also true. Then X and Y are **independent**.

$$(1) f_{XY}(x, y) = f_X(x) \cdot f_Y(y)$$

$$(2) f_{Y|x}(y) = f_Y(y) \text{ for all } x \text{ and } y \text{ with } f_X(x) > 0$$

$$(3) f_{X|y}(x) = f_X(x) \text{ for all } x \text{ and } y \text{ with } f_Y(y) > 0$$

$$(4) P(X \subset A, Y \subset B) = P(X \subset A) \cdot P(Y \subset B) \text{ for any sets } A \text{ and } B \text{ in the range of } X \text{ and } Y, \text{ respectively.}$$



Example: Machined Dimensions

- ♣ Let the random variables X and Y denote the **lengths of 2 dimensions** of a machined part.
- ♣ Assume that X and Y are independent and normally distributed. Find the desired probability.

	Normal Random Variables	
	X	Y
Mean	10.5	3.2
Variance	0.0025	0.0036

$$P(10.4 < X < 10.6, 3.15 < Y < 3.25) = P(10.4 < X < 10.6) \cdot P(3.15 < Y < 3.25)$$

=

=

=



Example: More Than Two Random Variables

- ♣ **Many dimensions** of a machined part are routinely measured during production. Let the random variables X_1 , X_2 , X_3 and X_4 denote the lengths of four dimensions of a part.
- ♣ What we have learned about joint, marginal and conditional PDFs in two variables extends to many (p) random variables.



Joint Probability Density Function Redefined

- ♣ The **joint probability density function** for the continuous random variables $X_1, X_2, X_3, \dots, X_p$, denoted as satisfies the following properties:

$$(1) f_{X_1 X_2 \dots X_p}(x_1, x_2, \dots, x_p) \geq 0$$

$$(2) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f_{X_1 X_2 \dots X_p}(x_1, x_2, \dots, x_p) dx_1 dx_2 \dots dx_p = 1$$

(3) For any region B of p-dimensional space,

$$P\left((X_1, X_2 \dots X_p) \subset B\right) = \iiint_B \dots \int f_{X_1 X_2 \dots X_p}(x_1, x_2, \dots, x_p) dx_1 dx_2 \dots dx_p$$



Mean & Variance of a Joint PDF

- ♣ The mean and variance of X_i can be determined from either the marginal PDF, or the joint PDF as follows:

$$E(X_i) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} x_i \cdot f_{X_1 X_2 \dots X_p}(x_1, x_2, \dots, x_p) dx_1 dx_2 \dots dx_p$$

and

$$V(X_i) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} (x_i - \mu_{X_i})^2 \cdot f_{X_1 X_2 \dots X_p}(x_1, x_2, \dots, x_p) dx_1 dx_2 \dots dx_p$$



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 - ◆ Poisson Process
- ♣ Continuous Random Variables
- ♣ Normal Approximation to the Binomial and Poisson Distributions
- ♣ More than One Random Variable and Independence
- ♣ **Functions of Random Variables**





Linear Functions of Random Variables

- ♣ A function of random variables is itself a random variable.
- ♣ A function of random variables can be formed by either linear or nonlinear relationships. We limit our discussion here to **linear functions**.
- ♣ Given random variables $X_1, X_2, \dots, X_p$ and constants

$$c_1, c_2, \dots, c_p$$

$$Y = c_1X_1 + c_2X_2 + \dots + c_pX_p$$

is a linear combination of $X_1, X_2, \dots, X_p$.



Mean & Variance of a Linear Function

♣ Let $Y = c_1X_1 + c_2X_2 + \dots + c_pX_p$

$$E(Y) = c_1E(X_1) + c_2E(X_2) + \dots + c_pE(X_p)$$

$$V(Y) = c_1^2V(X_1) + c_2^2V(X_2) + \dots + c_p^2V(X_p) + 2 \sum_{i < j} \sum c_i c_j \text{cov}(X_i X_j)$$

If $X_1, X_2, \dots, X_p$ are **independent**, then $\text{cov}(X_i X_j) = 0$,

$$V(Y) = c_1^2V(X_1) + c_2^2V(X_2) + \dots + c_p^2V(X_p)$$



Covariance Defined

The covariance between the random variables X and Y , denoted as $\text{cov}(X, Y)$ or σ_{XY} is

$$\sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)] = E(XY) - \mu_X\mu_Y$$

The units of σ_{XY} are units of X times units of Y .

For example, if the units of X are feet

and the units of Y are pounds,

the units of the covariance are foot–pounds.

Unlike the range of variance, $-\infty < \sigma_{XY} < \infty$.



Example

- ♣ Let X_i be a random variable with parameter p with $\mu = 1/p$ and $\sigma^2 = (1-p)/p^2$
- ♣ Let $Y = X_1 + X_2 + \dots + X_r$, a linear combination of r independent random variables.
- ♣ Then Y is a random variable with $\mu = r/p$ and $\sigma^2 = r(1-p)/p^2$.



Mean & Variance of an Average

$$\text{If } \bar{X} = \frac{(X_1 + X_2 + \dots + X_p)}{p} \text{ and } E(X_i) = \mu$$

$$\text{Then } E(\bar{X}) = \frac{p \cdot \mu}{p} = \mu$$

If the X_i are independent with $V(X_i) = \sigma^2$

$$\text{Then } V(\bar{X}) = \frac{p \cdot \sigma^2}{p^2} = \frac{\sigma^2}{p}$$



Reproductive Property of the Normal Distribution

If $X_1, X_2, \dots, X_p$ are **independent, normal** random variables

with $E(X_i) = \mu$, and $V(X_i) = \sigma^2$, for $i = 1, 2, \dots, p$,

then $Y = c_1X_1 + c_2X_2 + \dots + c_pX_p$

is a **normal** random variable with

$$E(Y) = c_1\mu_1 + c_2\mu_2 + \dots + c_p\mu_p$$

and

$$V(Y) = c_1^2\sigma_1^2 + c_2^2\sigma_2^2 + \dots + c_p^2\sigma_p^2$$



Example: Linear Function of Independent Normals

- ♣ Let the random variables X_1 and X_2 denote the independent length and width of a rectangular manufactured part. Their parameters are shown in the table.

	Parameters of	
	X_1	X_2
Mean	2	5
Std Dev	0.1	0.2

- ♣ What is the probability that the perimeter exceeds 14.5 cm?

Let $Y = 2X_1 + 2X_2 =$ perimeter



Exercise

♣ If X_1 and X_2 are independent random variables with $\mu_1 = 4$, $\mu_2 = 2$, $\sigma_1 = 3$, $\sigma_2 = 5$, and $Y = 4X_1 - 2X_2$, determine the following.

- $E(Y)$
- $V(Y)$
- $E(2Y)$
- $V(2Y)$